Adverse Selection in (Un)Subsidised Health Insurance: Evidence from Nepal's Age-70 Threshold*

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Governments in low- and middle-income countries are increasingly pursuing universal health coverage by building contributory health insurance programs in which enrolment mandates are weakly enforced and are therefore voluntary in practice. When enrolment is a choice, the premium determines both who enters the risk pool and the cost of insuring that pool. This paper presents the first quasi-experimental estimate of adverse selection in a national health insurance scheme in a low- or middle-income country. I exploit a statutory premium waiver at age 70 in Nepal's National Health Insurance Programme. Using the universe of enrolment and claims records, I implement a regression discontinuity design based on age at enrolment. Coverage nearly doubles at the threshold, increasing from 22 to 42 percent of the eligible population, and expands into areas that are poorer and farther from care. Claims per enrollee fall across every margin. Because enrolment itself responds to price, the discontinuity compares insurance pools with different compositions rather than a fixed population's healthcare use. The marginal enrollees induced by the zero price generate NPR 3,358 in claims per enrollee, compared with NPR 5,269 among those already paying, yielding a ratio of 0.637. The paid pool is therefore adversely selected, despite household bundling and restricted enrolment windows.

Keywords: Adverse Selection, Health Insurance, Regression Discontinuity, Nepal

JEL-Classification: D82, G22, H51, I13, I18, O15

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1 — Introduction

Governments in low- and middle-income countries have significantly increased public health financing over the past twenty years. This expansion aims to achieve universal health coverage (UHC), mainly through national insurance programmes rather than relying solely on the health budget (Lagomarsino et al. 2012; Kurowski et al. 2023; Watson, Yazbeck, and Hartel 2021). Health insurance programmes are attractive to these governments because they protect households against the financial risks associated with health shocks while allowing governments to collect premiums that would otherwise have been paid out-of-pocket (Das and Do 2023; Chetty and Looney 2006). Since much of the population remain outside the tax net, or formal payroll from which a contribution could be withheld, these schemes operate as voluntary programmes in practice (Banerjee et al. 2021; International Labour Organization 2023). Countries like Ghana, Kenya, the Philippines, Vietnam and Indonesia each have a contributory scheme which is compulsory by law and collects premiums directly from those it enrols (Lagomarsino et al. 2012), and Nepal's programme, the setting of this paper, is built the same way.

Voluntary enrolment therefore raises a fundamental policy concern regarding the composition of the risk pool. When coverage is not mandatory, take-up may be limited and disproportionately concentrated among individuals who anticipate greater healthcare utilisation (Akerlof 1970; Rothschild and Stiglitz 1976). As a result, enrolment remains low, while the pool of enrollees incurs higher costs per member than the population from which it is drawn. This undermines the fiscal rationale for voluntary coverage: individuals have proved unwilling to pay unsubsidised premiums (Das and Do 2023), leaving governments to finance the resulting gap through general revenue, subsidising poorer households and covering deficits as they arise. The extent to which a scheme can expand coverage while remaining financially viable therefore depends in part on the degree to which premiums sort individuals by risk. Adverse selection is thus not merely a question of how much enrolment can be increased, but also of whether such schemes can remain financially viable.[1]

1. Nepal's programme has already showed financial challenges. Payments owed to empanelled hospitals was about NPR 16 billion by early 2026 and more than fifty hospitals partially suspended acceptance of insurance cards by early 2026. Health Insurance Board also responded by capping outpatient coverage to NPR 25,000 and introducing policies of co-payment for medicines. (The Kathmandu Post 2026a, 2026b).

Evidence on adverse selection in health insurance markets in low- and middle-income countries remains limited and mixed. While some studies find evidence of adverse selection, with individuals in poorer health more likely to enrol in insurance (Fischer, Frölich, and Landmann 2023; Yao, Schmit, and Sydnor 2017; Asuming, Kim, and Sim 2024), others find little or no evidence of selection (Wagstaff et al. 2016; Hillebrecht et al. 2021; Banerjee, Duflo, and Hornbeck 2014). This divergence partly reflects the empirical difficulty of identifying selection. The standard test examines whether insurance coverage is positively correlated with realised risk (Chiappori and Salanié 2000). While such a correlation provides evidence that selection is present, it does not establish either its direction or magnitude. Identifying these requires variation in the price of coverage that changes enrolment decisions without simultaneously affecting individuals' underlying health (Einav and Finkelstein 2011). National insurance schemes rarely provide such variation outside of experimental settings. Where adverse selection has been identified two design remedies are suggested. Bundling coverage so that insurance cannot be purchased for individual members alone (Fischer, Frölich, and Landmann 2023), and restricting enrolment to specified periods, thereby preventing individuals from purchasing coverage only when they anticipate needing it, as shown by Banerjee et al. (2021).

Nepal's National Health Insurance Programme (NHIP) incorporates both of these design remedies while also providing a source of exogenous price variation.[2] The programme is contributory and, in practice, voluntary. A household of up to five members pays an annual premium of NPR 3,500 (USD 26), with coverage bundled across household members. Enrolment is restricted to four fixed start dates each year and must be arranged one quarter in advance, preventing households from waiting until healthcare is needed before purchasing coverage. Since 2019, the premium has been waived for individuals aged 70 or above at the beginning of an enrolment spell. The programme therefore provides a discrete reduction in the price of coverage to zero at an age determined by programme rules, rather than by an individual's health status. If the enrollees induced by the zero price subsequently claim less than those who remained enrolled when the positive premium applied, this would imply that the premium had screened out lower-cost individuals while retaining higher-cost ones (Einav and Finkelstein 2011).

2. Subedi (2026) provides a detailed account of the programme's design, implementation and its effects on utilisation.

I estimate this comparison using a local linear regression discontinuity design around age 70 at enrolment, comparing enrolment spells that begin shortly before an enrollee's seventieth birthday with those that begin shortly after. I construct a dataset from the universe of enrolment records held by the Health Insurance Board and the reimbursement claims submitted to it. I geocode enrollees' wards and empanelled facilities to measure each enrollee's distance to care and the number of empanelled facilities within 10 km of their ward. To my knowledge, this is the first study to assemble a dataset of this kind. Because the data cover only individuals who enrol, the parameter of interest requires careful interpretation. The design compares the composition of the pool enrolling at a positive premium just below age 70 with that enrolling at a zero premium just above age 70. It therefore identifies a change in pool composition, rather than the effect of free insurance on healthcare utilisation for a fixed population. The enrolled population itself changes discontinuously at the threshold, and this re-sorting is the object of interest. In a voluntary insurance market, a price change operates by changing who chooses to buy coverage; the discontinuity therefore captures how this re-sorting changes the risk borne by the programme and, consequently, its cost per enrollee.

I have three main findings. First, coverage increases sharply at the age-70 threshold, rising from 22 percent of the age-eligible population just below age 70 to 42 percent just above it. The waiver therefore induces an additional one-fifth of the eligible population to enrol. The enrollees above the cutoff also differ systematically in their access to care and local economic conditions: those enrolling at the threshold live 0.54 km farther from their assigned first point of contact, have 0.74 fewer empanelled facilities within 10 km of their ward, and reside in poorer local units. The waiver therefore expands coverage disproportionately into areas where access to care and local resources are more limited.

Second, claims per enrollee decline across the same threshold. Total claims fall by NPR 921.5, a 17.4 percent reduction; the probability of making any claim falls by 5.9 percentage points, or 11.8 percent; and the mean claim amount falls by NPR 161.6, or 15.0 percent. Inpatient care exhibits the largest proportional decline: the number of inpatient claims falls by 19.4 percent, while the probability of an inpatient stay declines by 17.0 percent. Because inpatient stays cannot be initiated at the enrollee's discretion in the same way as some other forms of care, these results are less readily explained by differences in discretionary utilisation. I show that the findings are robust to alternative bandwidths, donut specifications, polynomial orders, and kernel weights, and that the corresponding effects are absent at placebo thresholds.

Third, I estimate the marginal cost of the enrollees induced by the waiver by adapting the framework of Einav, Finkelstein, and Cullen (2010) to the regression discontinuity design in similar spirit to Finkelstein, Hendren, and Shepard (2019). Under the assumptions described in Section 5, the marginal enrollees drawn into the programme by the waiver have an estimated cost of NPR 3,358 each in claims, compared with NPR 5,269 for the pool that was already paying the premium, yielding a marginal-to-incumbent cost ratio of 0.637. This estimate is close to the nearest quasi-experimental benchmark, a premium discontinuity in the Massachusetts subsidised exchange, where the corresponding ratio is 0.59.[3]

This paper makes three contributions. First, to my knowledge, it provides the first quasi-experimental estimate of adverse selection in a national health insurance scheme in a low- or middle-income country, identified from cost curve rather than from a positive correlation test. The existing estimates are drawn predominantly from high-income settings (Einav, Finkelstein, and Cullen 2010; Hackmann, Kolstad, and Kowalski 2015; Panhans 2019; Finkelstein, Hendren, and Shepard 2019). The closest exception is Fischer, Frölich, and Landmann (2023), who identify adverse selection in a low-income insurance market in Pakistan using a randomised experiment on a micro-insurer, rather than a price change implemented across a national scheme. Second, the paper contributes to the literature on take-up of subsidised health insurance in low- and middle-income countries (Thornton et al. 2010; Wagstaff et al. 2016; Banerjee et al. 2021). Whereas this literature relies almost entirely on experimental evidence, I estimate the enrolment response to a statutory price reduction in a national insurance scheme. By geocoding both individuals and empanelled facilities, this paper also contributes to the literature on who public insurance reaches. It characterizes the enrollees induced by a zero price in terms of both their access to care and the poverty of the areas in which they live, extending the focus beyond the utilization gaps documented by Dupas and Jain (2024). Third, the paper contributes to the literature on design-based mechanisms for mitigating adverse selection by examining these mechanisms in a setting where they are already in force (Fischer, Frölich, and Landmann 2023; Banerjee et al. 2021). NHIP bundles coverage at the household level and restricts enrolment to fixed windows, both of

3. Constructed from the estimates of Finkelstein, Hendren, and Shepard (2019) rather than reported by them.

which apply below age 70. Despite these safeguards, the pool enrolled under the positive premium remains adversely selected (Section 5).

1.1. Related Literature

Private information about risk is a central concern in insurance markets. When individuals know more about their expected costs than insurers, premiums can induce sorting that changes the composition of the insured pool (Arrow 1963; Akerlof 1970; Rothschild and Stiglitz 1976). Empirically, this selection need not be adverse: individuals may differ in both expected costs and willingness to purchase insurance, so selection on preferences can offset or reverse selection on risk (Finkelstein and McGarry 2006; Fang, Keane, and Silverman 2008). Selection may also operate through expected behavioural responses to insurance, with individuals who anticipate greater utilisation sorting into more generous coverage (Einav et al. 2013). Insurance can also change the behaviour of individuals after enrolment, generating moral hazard (Aron-Dine, Einav, and Finkelstein 2013; Einav and Finkelstein 2018). Selection is concerned with changes in the composition of individuals who hold coverage, whereas moral hazard is concerned with changes in utilisation among individuals whose coverage changes. This distinction is important and determines the estimand in this paper because the data is conditional on enrolment. The direction and magnitude of selection are therefore empirical questions.

The first-generation approach tests whether coverage is positively correlated with realised risk (Chiappori and Salanié 2000). This establishes the presence of selection but does not determine its direction or magnitude, and may yield a null result when selection on preferences offsets selection on risk (Finkelstein and McGarry 2006). Recent empirical work studies selection through the relationship between insurance prices, enrolment, and costs. Einav, Finkelstein, and Cullen (2010) show how price variation can be used to estimate a cost curve describing the expected cost of individuals induced to enrol as the price changes. A downward-sloping marginal cost curve indicates adverse selection: individuals induced to enrol by a lower price have lower costs than those already enrolled. This approach provides a direct measure of the economic consequences of selection and can be combined with demand to conduct welfare analysis (Einav and Finkelstein 2011). A growing empirical literature applies this framework to health insurance, using price variation generated by employer contributions, policy reforms, and discontinuities in subsidies or premiums (Einav, Finkelstein, and Cullen 2010; Hackmann, Kolstad, and Kowalski 2015; Panhans 2019; Finkelstein, Hendren, and Shepard 2019). This literature is concentrated in high-income settings with private or contributory insurance markets, where individuals typically choose among competing contracts or face different prices for coverage. The implications for national public insurance schemes are less clear. In a national scheme with a common benefit package and a statutory premium, price variation need not induce sorting across competing contracts; instead, it changes whether otherwise eligible individuals enrol at all.

Evidence on selection and price-induced enrolment in low- and middle-income countries is limited and mixed. Studies of micro-insurance and subsidised insurance find evidence of both adverse and weak or absent selection (Yao, Schmit, and Sydnor 2017; Fischer, Frölich, and Landmann 2023; Asuming, Kim, and Sim 2024; Wagstaff et al. 2016; Banerjee, Duflo, and Hornbeck 2014; Hillebrecht et al. 2021). The closest evidence to this setting is Banerjee et al. (2021), who show that temporary premium subsidies in Indonesia increased enrolment while attracting lower-cost individuals. Other work shows that enrolment design can mitigate selection: household-level bundling limits the ability to insure selectively (Fischer, Frölich, and Landmann 2023), while fixed enrolment windows can restrict enrolment immediately before anticipated healthcare use (Banerjee et al. 2021). Whether such mechanisms eliminate selection when they are embedded in a national scheme over the long run remains an open empirical question.

This paper contributes to this literature by studying a permanent statutory reduction in the price of national health insurance from a positive premium to zero. This setting provides quasi-experimental variation in the price of coverage while holding the insurance product fixed. Because the data contain only individuals who enrol, the design does not identify the effect of insurance on a fixed population. Instead, it identifies how the composition of the insured pool changes when the price falls to zero. This makes the design conceptually closer to the cost-curve literature (Panhans 2019; Finkelstein, Hendren, and Shepard 2019) than to age-based eligibility designs that estimate the effect of gaining coverage for a fixed population (Card, Dobkin, and Maestas 2008, 2009; Miller, Pinto, and Vera-Hernández 2013).

The rest of the paper proceeds as follows. Section 2 describes the programme and the data. Section 3 sets out the empirical design and tests of its validity. Section 4 reports the effects of the threshold on coverage, costs, and the health of enrollees. Section 5 maps these estimates into the selection framework. Section 6 presents the falsification tests and discusses threats to identification, and Section 7 concludes.

2 — Setting and Data

2.1. Nepal and the National Health Insurance Programme

Nepal is a lower-middle-income country with a population of 29.2 million (National Statistics Office 2021). Gross national income per capita was USD 1,380 in 2022 (World Bank 2025), while out-of-pocket payments account for more than half of total health expenditure, equivalent to USD 213 per capita (WHO 2022). Healthcare is delivered through a mixed public and private system organised across three tiers: primary health care centres, district hospitals, and tertiary hospitals. Reducing households' financial exposure to health shocks is therefore a central policy priority, with the National Health Insurance Programme (NHIP) serving as the government's flagship instrument for achieving this objective.

NHIP is a contributory public health insurance scheme launched in 2016 and administered by the Health Insurance Board (HIB) since 2017. Following an initial pilot in three districts, the programme expanded to all 77 districts by 2022 and covered approximately one-third of households by the end of that year (Subedi 2026). Enrolment occurs at the household level: households of up to five members pay an annual premium of NPR 3,500 (USD 26), with each additional member incurring a further NPR 700. In exchange, households receive cashless coverage subject to an annual benefit ceiling of NPR 100,000 (USD 750), shared across members. Covered services include preventive and curative care, inpatient and emergency treatment, surgery, diagnostics, and rehabilitation (HIB 2024). The programme also provides subsidised or free coverage to several groups. Female community health volunteers pay 50 percent of the standard premium, while ultra-poor households, individuals with disabilities, and patients with selected terminal diseases are fully exempt from premiums (HIB 2024).

Coverage is provided through a network of empanelled public and private facilities. Households enrol in person, after which each member receives an insurance card used to identify them at participating providers. At the point of care, insured members present their card and face no direct payment: throughout the study period, NHIP operates on a fully cashless basis, with neither copayments nor deductibles. Instead, the treating facility submits the claim to the HIB and receives reimbursement directly according to pre-specified rates covering the visit, physician fees, diagnostics, medicines, and other services included in the benefit package. NHIP operates as a single public insurance system, with the HIB responsible for enrolment, provider empanelment, and reimbursement through a central information system. This system generates the administrative enrolment and claims records used in this paper.

2.2. The Age-70 Discontinuity

Since April 2019, NHIP has waived premiums for citizens aged 70 and above and provided each eligible individual with a separate benefit ceiling of NPR 100,000, independent of the allocation available to their household. Table 1 summarises the change in insurance terms at the age-70 threshold. Two policy changes occur simultaneously: the price of insurance falls to zero, and the unit of coverage shifts from the household to the individual. At no age can enrollees choose a more or less generous insurance plan, as the benefit package is uniform on both sides of the threshold. The discontinuity therefore reflects a change in price and coverage unit, rather than a change in the quantity of insurance chosen.

Table 1 — Insurance Terms Below and Above Age 70

	Below 70	70 and above
Annual premium	NPR 3,500 (USD 26)	Free
Benefit ceiling	NPR 100,000 per household	NPR 100,000 per individual
Coverage unit	Household	Individual

Note: Premium of NPR 3,500 applies to households of up to five members; each additional member costs NPR 700. At age 70 the individual leaves the household plan and receives a separate free individual policy.

The household premium structure means that the budgetary saving associated with reaching age 70 varies with household composition and cannot be characterised by a simple per-person average. For a member of a single-person household, turning 70 eliminates the household's entire premium obligation, generating a marginal saving of NPR 3,500. By contrast, when a member of a household with two to five members reaches age 70, the remaining members continue to owe the NPR 3,500 household premium, so the marginal saving to the household is approximately zero. Only in households with more than five members does the removal of a 70-year-old reduce the premium by the NPR 700 charge for an additional member. Although the average premium per person falls from NPR 3,500 to NPR 700 as household size increases from one to five, this average does not represent the marginal price saving associated with one member crossing the age-70 threshold.The analysis that follows does not require choosing between these measures of the price change. The selection analysis in Section 5 is identified solely from coverage and realised costs, and therefore does not enter the premium schedule into the calculation. The empirical design requires only that the price of coverage falls to zero at age 70 for all eligible individuals, regardless of the size of the premium saving realised by their household.

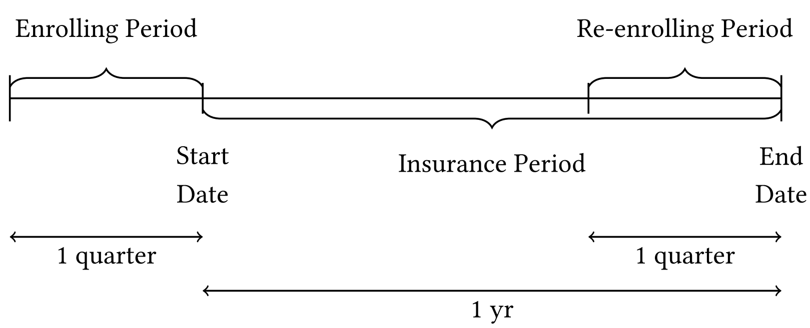
For the individual crossing the threshold, the change in insurance terms is unambiguous: her share of the premium obligation lapses, even when the household's total premium remains unchanged, as in households with two to five members. At the same time, she receives an individual NPR 100,000 benefit entitlement that is no longer shared with or competed for by other household members. Whether changes in price or in the unit of coverage drive enrolment and utilisation is therefore an empirical question. In practice, however, benefit ceilings are unlikely to bind for most enrollees: fewer than 0.1 percent of enrolment spells on either side of the threshold reach the NPR 100,000 annual ceiling (Table C2), while the 90th percentile of claims is NPR 20,317 among household-spells below age 70. The change in the coverage unit at age 70 therefore has limited scope to affect behaviour. Section 5.4 considers this issue as a condition for interpreting the cost comparison.

2.3. Enrolment Mechanics

NHIP operates on a quarterly enrolment cycle (Figure 1). Insurance spells begin on one of four dates each year—February, May, August, or November—with registration opening one quarter before each start date. Renewals are permitted only during the quarter preceding the expiration of the current spell, with the subsequent spell beginning on the day after the previous one ends. This structure does more than prevent re-enrolment during an existing spell. Because coverage can begin only on a fixed date and enrolment must be completed a quarter in advance, individuals cannot wait until they anticipate needing care and then obtain immediate coverage. Restricting the timing of enrolment in this way is therefore one of the programme's mechanisms for limiting adverse selection.

Eligibility for the premium waiver is determined by an individual's age on the spell's *start date*: those aged 70 or above when their spell begins pay no premium. Age at the start date, measured in days, therefore serves as the running variable in the empirical design. This enrolment-timing rule constitutes one of NHIP's two safeguards against adverse selection, the other being household-level bundling. Both mechanisms apply below the age-70 threshold, yet, as shown in Section 5, the pool insured under these safeguards remains adversely selected.

Figure 1 — NHIP Enrolment Timeline



2.4. Data

2.4.1. Administrative data. I use administrative data from the Health Insurance Board covering the universe of NHIP enrollees and reimbursed claims from the programme's inception through May 2025. The data comprise two linked files. The registration file records each enrolment spell and contains individual- and household-level identifiers, date of birth, gender, enrolment and insurance-period dates, premium paid, registration category, and first point-of-contact facility. The claims file records each reimbursed service, including the claim date, amount, ICD-10 diagnosis code, hospital, and service type.

The unit of analysis is the enrolment spell. The analysis sample comprises all spells with start dates between May 2023 and May 2024, a period selected to ensure that the programme's rules and institutional environment remain constant throughout the sample. By 2022, NHIP had expanded to all 77 districts, so this cohort is drawn from a nationally complete programme rather than one still undergoing geographic expansion. Benefit rules are unchanged over the sample period, and care is fully cashless: copayments were introduced only in 2025, after the sample period ends. The claims file records reimbursements rather than services, meaning that a claim enters the data only after reimbursement has occurred. Because reimbursement typically lags the underlying service by several months, spells ending later in the sample are observed less completely than those ending earlier, leading to an understatement of the level of claims. This does not bias the discontinuity, however, because all spells within an enrolment cohort share the same start date and therefore the same claims observation window; the resulting incompleteness thus affects both sides of the threshold symmetrically. And I report the ratio of marginal to average cost in Section 5 because it is invariant to the common proportional understatement of claims if any.

The single-year sample also means that each individual enters the analysis only once, making the design a cross-section rather than a panel of repeated enrolment spells. For each spell, I aggregate claims into spell-level outcomes: an indicator for any claim, the number of claims, total claims, and the mean claim amount. Enrollees with no claims during their spell are assigned zero for the relevant outcomes. Within the 485-day window around the seventieth birthday used to estimate total claims (Section 3.1), the sample contains 127,359 spells. Just below the threshold, claims are relatively common but modest: 49.8 percent of spells record at least one claim, the average spell generates 2.52 claims, and total claims average NPR 5,310 per spell. The mean claim amount, assigning zero to non-claimers, is NPR 1,076. Table 2 reports summary statistics for the estimation sample on both sides of the threshold.

2.4.2. Census data. I supplement the administrative records with data from the 2021 Nepal Census. The administrative data identify who enrols, whereas the census characterises the population from which enrollees are drawn, regardless of insurance status. Because Table 2 conditions on enrolment, differences between the two sides of the threshold are expected: the waiver changes who chooses to enrol, and this change in composition is precisely the outcome of interest. The identifying assumption instead requires the underlying eligible population to be continuous at the cutoff (Finkelstein, Hendren, and Shepard 2019), which can only be assessed using data that also observe individuals without insurance. The census serves two purposes in this regard. First, it

Table 2 — Summary Statistics for the Estimation Sample

	Below 70		70 and above	
	Mean	SD	Mean	SD
Panel A. Claims outcomes				
Any claim	0.498	0.500	0.373	0.484
Number of claims	2.52	4.09	1.70	3.41
Total claimed (NPR)	5,310	11,081	3,779	9,674
Mean claim (NPR)	1,076	2,278	841	2,105
Panel B. Enrollee characteristics				
Female	0.519	0.500	0.520	0.500
Age (years)	68.8	0.4	70.3	0.4
Household size	5.23	2.02	2.64	2.46
Non-poor (full premium)	0.865	0.341	0.319	0.466
Ultra-poor (exempt)	0.023	0.151	0.011	0.102
Spells (N)	40,802		86,557	

Note: The sample is all enrolment spells starting May 2023–May 2024 within the 485-day window around the seventieth birthday, the MSE-optimal bandwidth for total claimed in Table 5, $N = 127{,}359$. Each observation is a spell; claims outcomes are zero for non-claimers. Household size counts members co-enrolled in the spell year and is formed before the age window is applied.

provides the covariates used in the balance test in Section 3.2. The administrative register does not record health or socioeconomics characteristics, whereas the census additionally records literacy, caste, religion, marital status, disability, economic activity, and household wealth. Second, it provides an independent test of continuity in the eligible population at age 70, reported in Appendix A.3.

Population denominators are drawn from United Nations single-year-of-age estimates for Nepal, which extend the census population estimates to the sample period (United Nations, Department of Economic and Social Affairs, Population Division 2024). Because the census is a November 2021 snapshot whereas the analysis sample covers 2023–2024, using census counts would mismeasure the population at risk by two years, particularly at ages where mortality is relatively high. The United Nations series interpolates population estimates by single year of age, and all population denominators used in the paper are derived from this series.[4]

4. A second reason to prefer the interpolated series is that the census records respondents' age rather than their date of birth, resulting in disproportionate heaping at multiples of five, including age 70. The United Nations series

2.5. *Distance and Poverty Incidence Data*

To measure distance to the FPOC, I geocode each empanelled facility using a combination of the Google Maps API and manual searches for unmatched facilities. I geocode each enrollee's ward of residence using Nepal's administrative boundary shapefile and assign each individual to the centroid of their ward. I then calculate the Haversine distance between the ward centroid and the enrollee's assigned first point-of-contact facility. Only 1.2 percent of enrollment spells remain unmatched. For local poverty incidence, I use 2023 small-area estimates at the VDC level (National Statistics Office 2026).[5]

3 — Research Design and Validity

3.1. *Regression Discontinuity Design*

I exploit the age-70 premium waiver to estimate how a reduction in the price of insurance affects both demand for coverage and the cost of the pool that takes up coverage. The empirical design is a local linear regression discontinuity in age at enrolment, allowing both the intercept and the slope to vary on either side of the threshold. The unit of observation is the enrolment spell. I estimate

$$Y_i = \alpha + \tau D_i + \beta_0 A_i + \beta_1 D_i A_i + \varepsilon_i, \tag{1}$$

where Y_i denotes the spell-level outcome, A_i is age at the spell start date, measured in days and centred at the seventieth birthday, and $D_i = \mathbf{1}[A_i \geq 0]$ indicates eligibility for the premium waiver. The coefficients β_0 and β_1 allow a separate linear relationships between the outcome and age on either side of the cutoff. The coefficient τ captures the discontinuous change at age 70 and is the parameter of interest. Treatment is sharp: every spell beginning at age 70 or above receives the waiver, while no spell beginning below age 70 does. Standard errors are clustered at the household level, reflecting the level at which premiums are paid and enrolment decisions

does not exhibit this pattern. Age heaping therefore affects only the census microdata, where it can influence which individuals fall on either side of the cutoff in the balance test. Appendix A.1 describes the correction. It does not affect the running variable in the main design, which is calculated from date of birth recorded in the administrative register.

5. Poverty estimates are not available at the ward level, so I use VDC-level estimates. VDCs are one administrative level above wards, with each VDC comprising approximately nine wards.

are made. Appendix D.2 reports the same estimates with standard errors instead clustered on the value of the running variable.

Because the data cover only individuals who enrol, the choice of estimand is central to the interpretation of the design. The parameter τ captures the difference in claims outcomes between the pool enrolling at a positive premium just below age 70 and the pool enrolling at a zero premium just above the threshold.[6] It is therefore a *pool-composition contrast*, rather than the causal effect of free insurance on utilisation for a fixed population. The composition of the enrolled population itself changes discontinuously at the threshold: population coverage increases from 22 to 42 percent (Section 4). This compositional response is the object of interest. In a voluntary insurance market, a change in price operates precisely by changing who chooses to purchase coverage, so τ captures the effect of this re-sorting on the risk pool and, consequently, on the programme's cost per enrollee and budgetary sustainability.

Equation (1) deliberately contains neither covariates nor fixed effects. Since τ represents the total difference between the pools enrolling immediately below and above age 70, conditioning on characteristics that themselves change at the threshold would remove precisely the variation that identifies the effect of interest. The running variable is age in days, calculated directly from the date of birth recorded in the administrative register. There are 971 distinct running-variable values within the estimation window, so the continuity-based design does not require an adjustment for a discrete running variable (Kolesár and Rothe 2018).

I select the bandwidth separately for each outcome using the mean-squared-error-optimal criterion and report the robust bias-corrected inference proposed by Calonico, Cattaneo, and Titiunik (2014). For total claims, the outcome underlying the selection analysis in Section 5, the selected bandwidth is $h = 485$ days.

Within each bandwidth, I estimate local linear regressions using uniform weights, following Imbens and Lemieux (2008). The resulting estimator is therefore equivalent to ordinary least squares applied to spells within the selected bandwidth. Appendix D.1 examines the robustness of the results to alternative bandwidths, kernels, and polynomial orders.

The identification strategy requires the eligible population to evolve smoothly through age 70

6. The ultra-poor and other premium-exempt categories described in Section 2 face no change in price at age 70 and therefore remain in the estimation sample on both sides of the threshold.

(Finkelstein, Hendren, and Shepard 2019), such that the insurance contract is the only factor that changes discontinuously at the threshold. The main text evaluates this assumption in two ways: by testing covariate balance in the census (Section 3.2) and by estimating placebo discontinuities at ages without corresponding policy changes (Section 3.3). Two additional checks are reported in the appendix: a test of continuity in the density of the eligible population (Appendix A.3) and permutation-based inference checks (Appendix D.2).

3.2. *Covariate Balance at the Threshold*

The regression discontinuity design requires individuals just above and below age 70 to be comparable in characteristics other than their insurance subsidy. Testing this condition within the enrolled sample would be circular, since enrolment itself responds to the price change. I therefore assess balance using the 2021 Nepal Census, which covers the full population regardless of insurance status.

Census age is self-reported, and respondents disproportionately report ages at multiples of five. Age 70 is consequently both the policy threshold and a prominent heaping point. I address this problem in two ways. First, I exclude observations reporting age 70, estimating the discontinuity without using the heaped cell (Barreca, Lindo, and Waddell 2016; Barreca et al. 2011). This approach imposes no assumptions about which individuals rounded their age, but requires the treated-side intercept to be extrapolated from ages 71 and above. Second, I redistribute the excess mass at each heaped age across neighbouring ages in proportion to a smooth density estimated from non-heaped ages. I repeat this procedure $M = 50$ times and combine the resulting estimates using Rubin's rules (Heitjan and Rubin 1990; Rubin 1987), so that the reported standard errors incorporate both sampling variation and uncertainty arising from the heaping correction. Appendix A.1 describes the procedure in detail. Because the two approaches rely on different assumptions, consistency between them provides stronger evidence of balance than either approach alone.

Table 3 reports results for nine covariates covering demographic characteristics (sex, caste, religion, ever-married status, disability, and household size) and economic characteristics (economic inactivity, literacy, and wealth quintile). Under the heaping correction, none of the covariates exhibits a statistically significant discontinuity, with the smallest individual p-value equal to

0.26 for literacy. The covariates are also jointly balanced: a Wald test of the null that all nine discontinuities are zero, combined across the 50 imputed draws, yields $F(9, 9{,}941) = 0.65$ with $p = 0.751$ (Table A2). The donut specification reaches the same conclusion for seven of the nine covariates.

The two specifications differ for literacy and the share of individuals not working because of age. Under the donut specification, both estimates are between two and three percentage points higher and statistically significant at the 5 percent level. These differences do not provide evidence of a discontinuity in the underlying eligible population. Both covariates exhibit among the steepest age gradients, and excluding age 70 leaves a support that alternates between heaped and unheaped ages. A local linear fit over only three ages on each side can therefore generate an apparent discontinuity mechanically. Literacy provides the clearest illustration: its estimated discontinuity is $+0.023$ under the donut specification but -0.009 under the heaping correction. This reversal in sign is more consistent with residual heaping than with a genuine change at age 70. Consistent with this interpretation, applying the same procedure at ages without policy changes produces discontinuities in these two covariates that are larger than the age-70 estimates and alternate in sign. Appendix A.2 plots each covariate against age; none exhibits a visible step at age 70.

3.3. Placebo Cutoffs

If the discontinuity in claims at age 70 reflects the change in the insurance price rather than a general feature of healthcare utilisation at older ages, the same specification should reveal no discontinuity at ages where the insurance contract remains unchanged. I therefore re-centre equation (1) at each whole year of age from 64 to 76 and re-estimate the model, holding the bandwidth, kernel, and clustering fixed at the values used in the headline specification. Ages 69 and 71 overlap with control and treated group respectively so their bandwidth selection is limited to 365 days. Each placebo is estimated using spells from its own side of the policy threshold, ensuring that no placebo window overlaps the true cutoff. Figure 2 reports the resulting thirteen estimates. The exercise uses total claims, the outcome underlying the selection analysis in Section 5.

The only clear discontinuity occurs at age 70. Total claims fall by NPR 921.5 at the policy

Table 3 — Covariate Balance at Age 70: 2021 Nepal Census

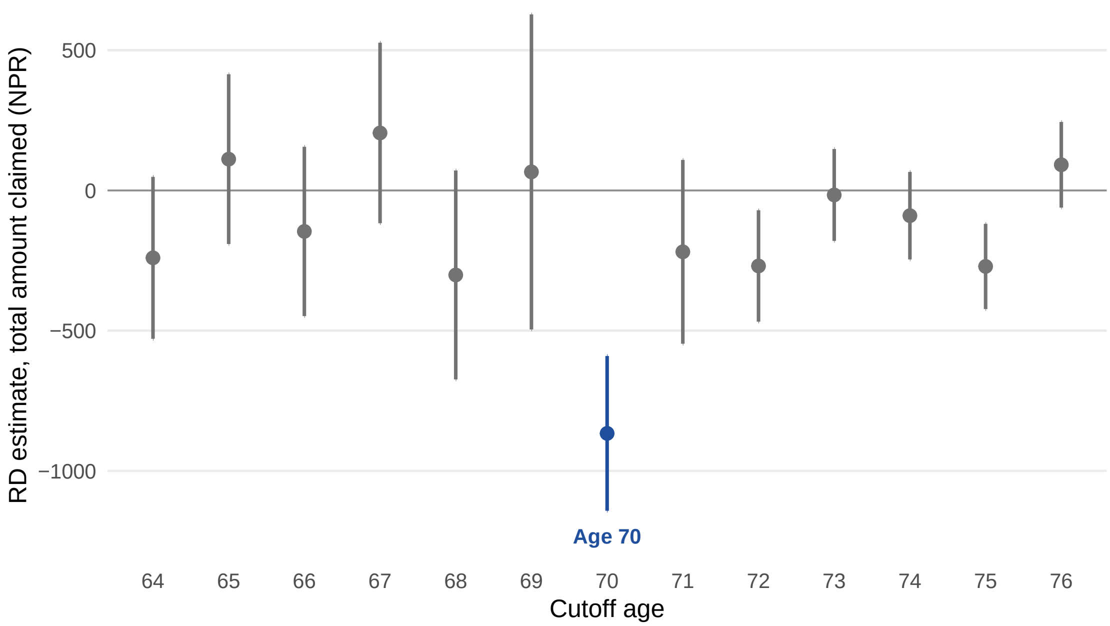
	Donut			Heaping-corrected		
Covariate	Mean	RD Est.	SE	Mean	RD Est.	SE
Female	0.505	−0.009	(0.011)	0.508	0.007	(0.009)
High Caste	0.476	−0.014	(0.011)	0.477	−0.009	(0.009)
Hindu	0.807	−0.008	(0.009)	0.810	0.004	(0.007)
Ever married	0.988	0.000	(0.002)	0.989	0.000	(0.002)
Has disability	0.064	0.003	(0.005)	0.064	−0.002	(0.004)
Household size	5.153	−0.111	(0.172)	5.203	0.121	(0.134)
Economically inactive (aged)	0.364	0.026^{**}	(0.011)	0.374	−0.001	(0.009)
Literate	0.341	0.023^{**}	(0.011)	0.328	−0.009	(0.008)
Wealth quintile	2.880	0.026	(0.036)	2.855	−0.024	(0.027)
Bandwidth (years)		3.65			3.41	
Observations		80,230			110,292	
Estimator	Local linear (OLS), uniform kernel					

Notes: Each cell is a local linear regression with a uniform kernel, weighted by the census individual weight, with standard errors clustered at the household level. Both bandwidths are selected on the sex indicator; because reported age is integer valued, both select the ages 67 to 73, the donut omitting 70. The columns differ in inference: donut p-values are the robust bias-corrected values of Calonico, Cattaneo, and Titiunik (2014), while the heaping-corrected p-values are conventional and pooled across the 50 draws. Mean Pre-70 is the weighted mean below the cutoff. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

threshold, corresponding to a 17.4 percent decline relative to the just-below-threshold mean of NPR 5,310. Across the 12 placebo ages, the estimated changes range from −NPR 301 to +NPR 205, equivalent to approximately 3–6 percent of the same mean, with the estimates alternating in sign rather than exhibiting a systematic age trend. The largest placebo estimate in absolute value is about third of the age-70 estimate, and the confidence interval at age 70 is distinct from those at the placebo thresholds.

Two placebo confidence intervals exclude zero, at ages 72 and 75. Both lie above the age-70 threshold, where the design has greater statistical power because the increase in coverage at 70 leaves the placebo specifications above the threshold with approximately twice as many spells as those below it. The resulting intervals are therefore tighter, making relatively small changes—of only a few percent of the mean—detectable above age 70 when comparable changes below the threshold would not be. Neither estimate is large. The placebo exercise is therefore more informative when viewed as a distribution of estimates rather than as a sequence of twelve independent hypothesis tests: the two significant placebo estimates are consistent with the scale of ordinary sampling variation rather than evidence of systematic discontinuities at non-policy ages. Nepal's old-age allowance begins at age 68 and is included in the sweep for completeness, although it lies outside the estimation window for total claims and therefore cannot generate a discontinuity at age 70.

Figure 2 — The discontinuity at different ages



Note: RD estimates for total claimed with equation (1) re-centred at each whole year of age, with robust bias-corrected inference proposed by Calonico, Cattaneo, and Titiunik (2014). Ages 69 and 71 have a bandwidth of 365 days so that their observations don't overlap control and treated side respectively.

3.4. *Interpreting the Discontinuity*

The premium waiver changes the composition of the enrolled pool through two channels. Some individuals would enrol at either price, in which case the waiver simply removes a premium they were willing to pay. Others enrol only when the price falls to zero, entering the pool above the threshold but not below it. The observed discontinuity therefore combines the utilisation response of the first group with the compositional response of the second: $\tau = \tau_S + \tau_M$, where τ_M captures the change in utilisation among individuals who would enrol at either price and τ_S captures the change in the composition of the enrolled pool. Enrolment type is not observed, so the data cannot separately identify these two components. The institutional features of NHIP, however, place strong restrictions on τ_M.

An enrollee's use of healthcare depends on the price faced at the point of care, rather than on the insurance premium already paid, which is sunk by the time care is consumed (Einav and Finkelstein 2018). During the study period, NHIP imposed neither copayments nor coinsurance, and the benefit package was identical on both sides of the age-70 threshold (Section 2). Care was therefore free at the margin for enrollees aged both 69 and 70, leaving little scope for ex-post moral hazard among individuals who would have enrolled at either price. Two channels remain. First, the unit of coverage changes from a household to an individual benefit ceiling. This can reduce the effective price of care in states where the household ceiling would otherwise bind, but the ceiling is slack for almost all enrollees on both sides of the threshold, making this channel quantitatively small. Second, the premium waiver transfers NPR 3,500 to the household, and if healthcare is a normal good, the resulting increase in disposable resources should weakly increase utilisation. Both channels therefore point towards a weakly positive τ_M, and both are likely to be small. I consequently treat τ_M as approximately zero and weakly positive.[7]

The compositional effect depends on a single object: the expected cost of the enrollees induced by the zero price relative to that of the pool that was already paying (Einav and Finkelstein 2011). If the marginal enrollees are cheaper, average claims decline as coverage expands, constituting the classical case of adverse selection (Akerlof 1970; Rothschild and Stiglitz 1976). If they are more expensive, average claims increase, corresponding to advantageous selection in which individuals who purchase coverage are more risk-averse and healthier (Finkelstein and McGarry 2006). Expected cost can differ across enrollees for two reasons: differences in underlying health and differences in the amount of care they would consume once insured. Individuals may therefore sort on either dimension. Einav et al. (2013) show that selection on the latter, often termed selection on moral hazard, is itself part of adverse selection rather than a separate force, because both mechanisms operate through expected costs. The theory therefore does not impose a sign on the compositional effect.

7. One channel works in the opposite direction. Enrollees below age 70 pay the premium whereas those above 70 do not, so a sunk-cost response could generate higher utilisation among paying enrollees and make τ_M negative.

The decomposition places a useful restriction on τ_S. When $\tau < 0$, $\tau_S = \tau - \tau_M$ must be at least as large in absolute value as τ, since $\tau_M \geq 0$ works against the observed decline. A fall in claims therefore provides a lower bound, in magnitude, on the compositional channel rather than representing the net effect of two opposing forces that could conceal a larger compositional response. If advantageous selection operates alongside classical adverse selection, it pushes τ_S in the opposite direction, implying that the underlying classical selection effect must be larger still.

The sign of the discontinuity is therefore informative in its own right. A decline in claims when coverage becomes free is consistent with the marginal enrollees being substantially lower-cost than the pool that was already insured. The result does not, however, establish the magnitude of the cost difference or directly price the resulting pool. Section 5 uses realised costs to estimate the cost of the marginal pool, while Section 6 examines alternative explanations for the decline.

4 — Results

4.1. Demand for Health Insurance

The first effect of the waiver is on demand for coverage. Figure 3 plots NHIP coverage—the share of the age-eligible population enrolled—by single year of age, counting an individual as insured if their enrolment spell covers 25 November 2023. Coverage remains broadly flat throughout the sixties before increasing sharply at age 70, from 22 percent immediately below the threshold to 42 percent immediately above it, corresponding to a discontinuity of $\Delta Q = 0.20$.[8] At the threshold, roughly one-fifth of the eligible population takes up coverage when the price falls to zero. The design therefore compares take-up at two administered prices rather than tracing a continuous demand curve between them.[9] This discontinuity in take-up is central to interpreting the subsequent results as changes in pool composition. The pool enrolled just above age 70 is not the same pool

8. The estimated increase is a lower bound. Enrolment operates through quarterly windows, and the premium is waived only from the first start date at which an individual is aged 70 or above. Consequently, the age-70 group includes individuals who have reached age 70 but have not yet reached an enrolment window at which the waiver applies.

9. The relevant price change is the statutory premium for a marginal individual policy. Below age 70, an individual whose household is not already enrolled can obtain coverage only by enrolling the household at NPR 3,500. At age 70, the individual can instead enrol alone at no charge.

observed just below the threshold with insurance made free; it is larger and differently selected. Consequently, each comparison across the threshold captures how the resulting re-sorting changes the characteristics and costs of the population insured by the programme.

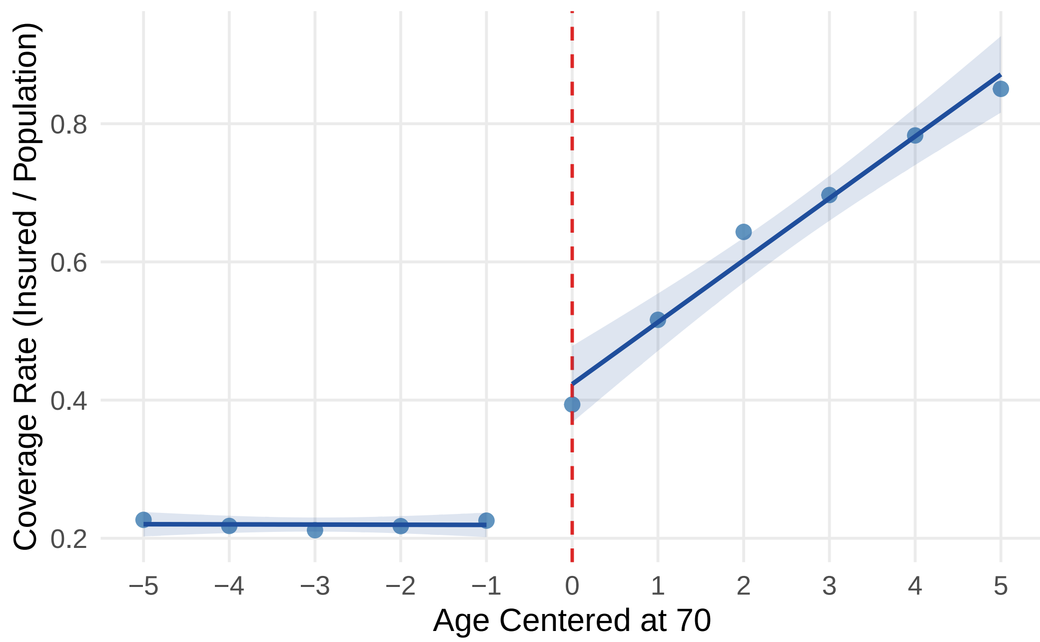
Table 4 examines who is induced to enrol by the zero price, reporting discontinuities in enrollee characteristics at age 70 rather than in utilisation outcomes.[10] The waiver therefore extends coverage beyond the population already willing to pay the premium. Enrollees at the threshold live 0.54 km farther from their assigned first point of contact, equivalent to 7.1 percent of the 7.57 km baseline among those enrolling just below the threshold. They also have access to 0.74 fewer empanelled facilities within 10 km of their ward, a 12.7 percent reduction relative to the baseline of 5.81. They are 3.0 percentage points less likely to reside in a municipality rather than a rural municipality, while the poverty rate of their local unit is 1.7 percentage points higher, representing a 10.1 percent increase relative to the below-threshold level of 0.163.

The composition of the enrolled pool also changes along characteristics associated with individuals rather than their locations. The share of enrollees belonging to high caste group (Khas-Arya Brahman and Chhetri) declines by 1.8 percentage points, equivalent to a 4.1 percent reduction relative to the below-threshold baseline of 0.428. At the same time, the share of female enrollees increases by 1.6 percentage points from a baseline of 0.519.[11] Both changes are smaller than the geographic differences, and neither is statistically significant at the 1 percent level. They are therefore better interpreted as consistent with the broader compositional pattern than as independent evidence of it. The programme is intended to extend insurance coverage to older Nepalis who would otherwise not purchase it, and the waiver clearly reaches this margin. The additional enrollees come disproportionately from poorer, more rural, and less well-served areas than those of individuals already willing to pay at age 69. In this sense, the waiver expands coverage precisely where the pre-existing coverage gap is greatest. Whether these newly insured individuals subsequently use the coverage is a separate question, addressed in Section 4.2.

10. Distance is measured to the assigned first point of contact, defined by the programme as the nearest public facility. The facility count is measured from the ward centroid and is therefore common to all individuals within a ward. Urban denotes any local unit other than a rural municipality. Local poverty is the headcount poverty rate of the enrollee's local unit.

11. Caste should be interpreted with some caution because it is recorded for only 80 percent of the sample, and the probability of recording itself changes discontinuously at the threshold. The result nevertheless reflects a change in composition rather than differential recording: characteristics observed for every enrollee are essentially unchanged when the analysis is repeated on the subsample with recorded caste.

Figure 3 — Coverage jumps at age 70



Note: NHIP enrollees as a share of the United Nations single-year population estimate for Nepal, measured at 25 November 2023 (United Nations, Department of Economic and Social Affairs, Population Division 2024). A person counts as insured if an enrolment spell covers that date. The dashed line marks age 70; fitted lines are local linear on each side of the threshold.

Table 4 — RD estimates at the age-70 threshold, by enrollee characteristic

	Individual				Local area	
	Distance to first contact (km)	Facilities within 10 km (count)	High caste (0/1)	Female (0/1)	Urban municipality (0/1)	Local poverty rate (share)
	(1)	(2)	(3)	(4)	(5)	(6)
RD estimate	0.536^{***}	-0.740^{***}	-0.0177^{*}	0.0155^{**}	-0.0303^{***}	0.0166^{***}
	(0.116)	(0.101)	(0.0070)	(0.0060)	(0.0052)	(0.0014)
R. p-value	<0.001	<0.001	0.055	0.016	<0.001	<0.001
Mean	7.57	5.81	0.428	0.519	0.756	0.163
% change	7.1%	−12.7%	−4.1%	3.0%	−4.0%	10.1%
BW (days)	485	485	485	485	485	485
N	125,859	127,359	101,502	127,359	126,601	127,359

Notes: Each observation is an enrolment spell. The running variable is the enrollee's age at the start date of the spell, measured in days and centred on the seventieth birthday. Estimates are local linear with separate slopes on each side of the cutoff and a uniform kernel, fitted by ordinary least squares with no covariates and no fixed effects. Every column is estimated at a common bandwidth of 485 days, the mean-squared-error-optimal choice of Calonico, Cattaneo, and Titiunik (2014) for claim value, so all columns rest on one window. Standard errors in parentheses are clustered at the household level; the reported p-value is the robust bias-corrected p-value. The estimation sample is 127,359 spells; columns 1, 3 and 5 are defined for 99, 80 and 99 percent of it respectively and every other column covers it in full. Local poverty is from the 2023 small-area estimation (National Statistics Office 2026). Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5 — RD estimates at the age-70 threshold, by claims margin

	Claim value		Had a claim			Number of claims		
	Value (NPR) (1)	Mean claim (NPR) (2)	Any (0/1) (3)	Chronic (0/1) (4)	Inpatient (0/1) (5)	Any (count) (6)	Chronic (count) (7)	Inpatient (count) (8)
RD estimate	-921.5^{***}	-161.6^{***}	-0.059^{***}	-0.048^{***}	-0.0128^{***}	-0.223^{***}	-0.144^{***}	-0.0204^{***}
	(126.9)	(21.9)	(0.007)	(0.007)	(0.0024)	(0.057)	(0.038)	(0.0039)
R. p-value	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	0.001	<0.001
Mean	5,310	1,081	0.50	0.36	0.075	2.56	1.39	0.105
% change	−17.4%	−15.0%	−11.8%	−13.2%	−17.0%	−8.7%	−10.4%	−19.4%
BW (days)	485	764	342	342	794	312	312	840
N	127,359	205,305	88,148	88,148	214,701	81,242	81,242	228,041

Notes: Each observation is an enrolment spell. The running variable is the enrollee's age at the start date of the spell, measured in days and centred on the seventieth birthday, so it is zero for a spell beginning on that birthday. Estimates are local linear with separate slopes on each side of the cutoff and a uniform kernel, fitted by ordinary least squares with no covariates and no fixed effects. Each bandwidth is the mean-squared-error-optimal choice of Calonico, Cattaneo, and Titiunik (2014) for the outcome, with one exception: the chronic column of each block is estimated at the bandwidth selected for its overall column, so columns 3 and 4, and columns 6 and 7, each rest on one sample and can be compared. Standard errors in parentheses are clustered at the household level; the reported p-value is the robust bias-corrected p-value. Every column is defined for the whole estimation sample, non-claimers included. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

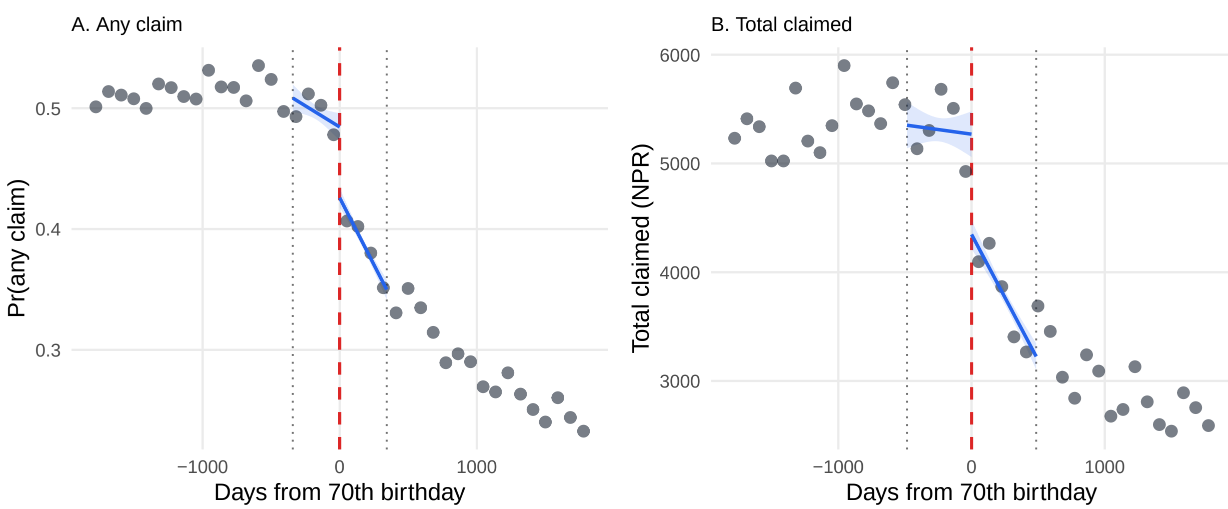
Table 5 reports claims accumulated over the twelve-month enrolment spell, comparing the pool that enrols when coverage is free with the pool that enrols while paying the premium. I focus on four sets of outcomes: total claim value; mean claim value; indicators for whether an individual has any claim, chronic claim, or inpatient claim; and the corresponding number of claims. Figure 4 presents the main discontinuities, while Appendix B.1 reports the remaining outcomes, which exhibit the same downward shift at the threshold.

Costs per enrollee decline across all four main margins. Total claims per spell fall by NPR 921.5 from a below-threshold mean of NPR 5,310, corresponding to a 17.4 percent decline. Mean claim value falls by NPR 161.6 from NPR 1,081, a 15.0 percent reduction (columns 1 and 2). The probability of recording any claim decreases by 5.9 percentage points from a below-threshold mean of 0.5, an 11.8 percent decline (column 3). The number of claims falls by 0.223 from 2.56 per spell, an 8.7

percent reduction (column 6). All four estimates are statistically significant at the 1 percent level. The pattern is therefore consistent across margins: fewer enrollees make claims, those who do claim make fewer claims, and the claims they make are smaller. The marginal enrollees brought into the programme by free coverage are consequently lower-cost than those who were already willing to pay.

The decline in claims could, in principle, reflect the fact that the waiver reaches enrollees who live farther from their first point of contact (FPOC), as shown in the previous section. Greater distance from care could itself reduce healthcare utilisation and therefore contribute to the observed decline in claims. I find little evidence that this mechanism drives the result. Appendix B.4 re-estimates the main effect within bands of distance from the enrollee's ward to their registered FPOC, and the estimated decline in claims is similar across distance bands, with no band differing significantly from another.

Figure 4 — The discontinuity in claims at age 70



Note: Binned means in quarterly bins of age at the spell start date, shown over twenty quarters on each side of the seventieth birthday, as in Figure 4 of the paper. The dashed vertical line is the cutoff; the dotted lines mark the estimation bandwidth, which is selected separately for each outcome and is 342 days in Panel A and 485 in Panel B. Local linear fits are estimated on each side of the cutoff on the spell-level data *inside the bandwidth only*, with a uniform kernel and no covariates or fixed effects, so the gap drawn at the cutoff is the coefficient reported in Table 5. Nothing is extrapolated: no fit is drawn beyond the bandwidth, and the bins outside it are shown for context and are descriptive only. Bins are a display device and no reported quantity depends on the bin width. Shaded bands are 95 percent confidence intervals from standard errors clustered on the household.

The composition of the pool also changes in when enrollees first use their coverage. Table B2 in the appendix shows that the probability of making a claim within 30 days falls by 4.9 percentage points from a below-threshold mean of 0.225, a 21.9 percent decline. The probability of making a claim within 90 days falls by 5.7 percentage points from 0.369, a 15.5 percent decline. Both reductions are proportionally larger than the decline in the probability of making any claim over the full spell, indicating that the threshold disproportionately excludes enrollees who would have claimed early. Individuals who enrol because they anticipate needing care are more likely to use their coverage soon after enrolment. Consistent with this, among enrollees who make a claim, the time from the start of the spell to the first claim increases by 7.4 days from a mean of 67.7 days, a 10.9 percent increase. Because this measure conditions on making a claim, the claimant pool itself changes at the threshold, so the estimate is descriptive rather than a causal effect on the timing of care.

4.3. The Health of the Enrollees

The enrollees who enter the programme when coverage is free are not only lower-cost on average; they also appear to be healthier. I examine this along two margins: treatment for chronic conditions (columns 4 and 7) and hospital admission (columns 5 and 8). Figure 5 plots the corresponding discontinuities.

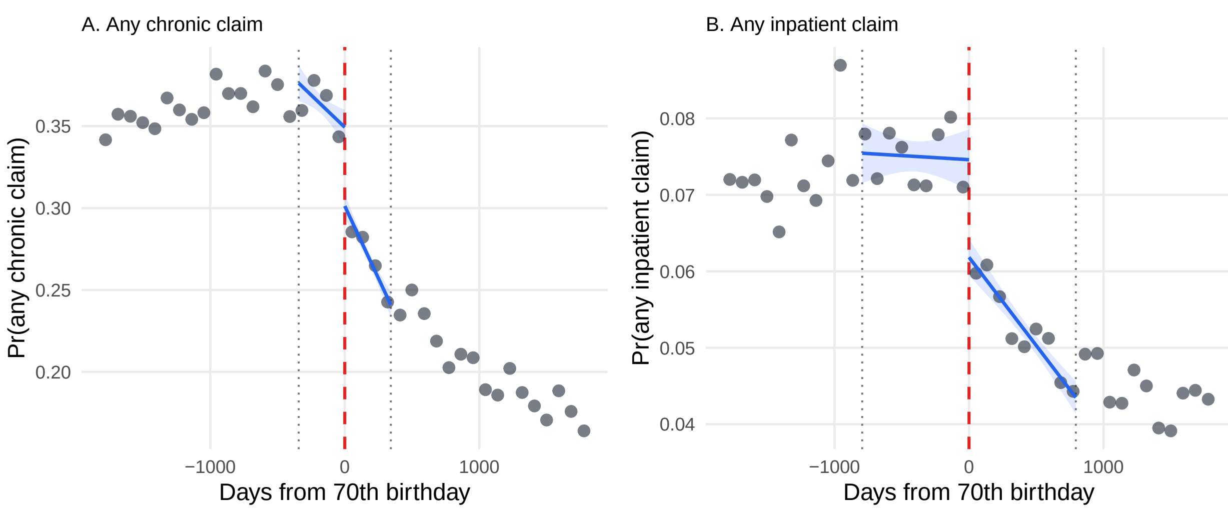
Chronic conditions provide the closest available proxy in the claims data for a predetermined health state, since conditions such as diabetes are not plausibly caused by the premium waiver. I classify claims as chronic using their ICD-10 codes, following Banerjee et al. (2021).[12] Chronic conditions account for 53.7 percent of claims in the sample, with hypertension and type-2 diabetes alone accounting for 15.0 and 10.0 percent, respectively. This composition is consistent with the healthcare needs of an older population. Chronic claims decline by more than claims overall: the probability of a chronic claim falls by 13.2 percent relative to a below-threshold mean of

12. The classification is based on the Chronic Condition Indicator developed by the Healthcare Cost and Utilization Project, which classifies each ICD-10-CM diagnosis code as chronic or non-chronic (Agency for Healthcare Research and Quality 2018). The indicator is then aggregated to the three-digit code stem, with a diagnosis classified as chronic when more than 75 percent of codes within its stem are chronic. The stem is used because it is the finest level at which ICD-10-CM and the World Health Organization's ICD-10 correspond. The underlying classification is based on the US clinical modification, whereas NHIP records diagnoses using the WHO classification; the same crosswalk therefore applies here as in Banerjee et al. (2021). Of the 1,910 stems in the indicator, 583 satisfy the 75 percent threshold. I use the 2018 version cited by Banerjee et al. (2021), which classifies codes as either chronic or non-chronic. The indicator has since been revised to include a third category for codes with no determination, which would alter the denominator of the 75 percent rule.

0.36 (column 4), while the number of chronic claims falls by 10.4 percent relative to an overall decline of 8.7 percent (column 7). Chronic claims account for 54.6 percent of claims below the threshold but 64.7 percent of the decline at the threshold. The reduction in utilisation is therefore disproportionately concentrated in the management of ongoing conditions.

Hospital admissions show a similar pattern. An inpatient admission reflects a clinical decision threshold that an outpatient visit need not, and is not plausibly something an enrollee can initiate at will. The probability of an inpatient claim falls by 1.3 percentage points from a below-threshold mean of 0.075, corresponding to a 17.0 percent decline (column 5). The number of inpatient claims falls by 0.020 from a mean of 0.105, a 19.4 percent reduction (column 8). Both estimates are statistically significant at well below the 1 percent level. The number of inpatient claims shows the largest proportional decline, exceeding the 17.4 percent reduction in total claims. The probability of an inpatient claim falls by a similar 17.0 percent, with both declines substantially larger than the 11.8 percent reduction in the probability of making any claim.

Figure 5 — The discontinuity on the two health margins



Note: Constructed as Figure 4. Panel A is drawn at the bandwidth selected for any claim, 342 days, rather than at its own, which is the pairing rule of Table 5; Panel B is drawn at its own bandwidth of 794 days, so the two panels display the same twenty quarters of data but are estimated on different windows. The gap drawn at the cutoff is the coefficient reported in columns 4 and 5 of that table.

The pool that pays is therefore in worse health on two margins for which moral hazard is unlikely to play a major role. Neither measure is a baseline indicator of health: both are based on diagnoses recorded when care is sought and are therefore post-treatment outcomes. I consequently treat them as components of the utilisation response rather than as controls or balance-test covariates. The pattern is consistent with adverse selection, but does not by itself establish its presence.

5 — Adverse Selection at the Age-70 Threshold

Section 4.2 showed that claims per enrollee fall at the threshold, while Section 3.4 showed that this decline reflects a change in the composition of the enrolled pool rather than a change in utilisation among individuals who would enrol at either price. What remains unknown is the cost of the enrollees induced by the waiver. They are not observed as a separate group: above the threshold, they are part of an average that also contains individuals who would have enrolled at either price. Identifying selection therefore requires recovering the costs of these two groups separately. I use the waiver as a source of variation in the composition of the insured pool to estimate the cost of the enrollees it induces and compare it with the cost of those who would enrol while paying the premium.

I adapt the framework of Einav, Finkelstein, and Cullen (2010) to make this comparison. NHIP offers a single contract defined by statute, leaving eligible individuals with a binary choice between enrolment and non-enrolment. Selection therefore operates through participation—the margin described by Akerlof (1970)—rather than through sorting across a menu of coverage levels (Rothschild and Stiglitz 1976).[13] Finkelstein, Hendren, and Shepard (2019) apply the same approach to premium discontinuities in the Massachusetts subsidised exchange, estimating the cost of the enrollees induced by each price change. I apply the framework to a single discontinuity. At age 70, the insurance price falls to zero and the unit of the benefit ceiling changes, but the number of contracts available does not. I treat the change in the benefit ceiling as a condition on the interpretation of the calculation in Section 5.4.

The estimates indicate adverse selection among those who enrol under the paid regime. The enrollees induced by the waiver—roughly one-fifth of the age-eligible population—generate NPR 3,358 in paid claims per enrollee, compared with NPR 5,269 among the enrollees retained by

13. The single-contract setting also limits what this exercise can capture. Selection may affect which contracts insurers choose to offer, a margin that Einav and Finkelstein (2011) argue can account for a substantial share of the welfare consequences of selection. In NHIP, however, the contract is fixed by statute rather than determined by competing insurers, so this margin is absent rather than assumed away.

the paid regime, yielding a ratio of 0.637. The marginal enrollees are therefore approximately 36 percent cheaper to cover than the pool they join. This is the same configuration that Finkelstein, Hendren, and Shepard (2019) interpret as evidence of adverse selection at premium discontinuities.

5.1. *Setup and Assumptions*

Consider a population of eligible individuals who may purchase a single insurance contract at price p. Index individuals by i, and let w_i denote their willingness to pay for the contract and c_i the expected cost of insuring them. Demand at price p is the share of individuals with $w_i \geq p$, denoted $D(p) = \Pr(w_i \geq p)$, and I write $Q = D(p)$ for the coverage rate. Ordering the eligible population by willingness to pay allows both cost curves to be expressed as functions of coverage. $MC(Q)$ denotes the expected cost of the marginal individual at coverage rate Q, that is, the individual induced to enrol at that coverage level, while $AC(Q)$ denotes the average cost of all individuals insured at that rate. By construction, the two are related through the accumulation of costs across the insured pool:

$$AC(Q)\,Q \;=\; \int_0^Q MC(q)\,\mathrm{d}q, \qquad \text{so} \qquad AC'(Q) \;=\; \frac{MC(Q) - AC(Q)}{Q}. \tag{2}$$

Average cost therefore falls whenever the cost of the marginal enrollee lies below the average cost of the existing pool.

The argument rests on following assumptions.

Assumption 1 (Monotonicity): *No one who enrols at the statutory premium would decline coverage when the price falls to zero.*

Assumption 2 (Continuity): *The eligible population is continuous at the threshold, so that the pools on either side are drawn from the same underlying population.*

Assumption 3 (Cost): *The cost of insuring an individual is measured by the claims paid by the programme on their behalf.*

Assumption 4 (No Differential Behavioural Response): *Crossing the threshold does not change how an individual who would enrol at either price uses care, so a change in measured cost reflects a change in the composition of the insured pool.*

Assumption 3 follows Finkelstein, Hendren, and Shepard (2019), who define insurer cost as medical claims paid and abstract from administrative costs.[14] Together with the contract-equivalence condition established in Section 5.4, these assumptions are sufficient for the calculation in the next subsection. Assumption 4 is particularly important: without it, the difference in total costs across the threshold would combine the cost of the individuals induced to enrol with a change in utilisation among those who would have been insured at either price.

5.2. *What the Waiver Identifies*

The waiver identifies the insured share and the average cost of the insured pool on either side of the threshold. Below the threshold, where coverage requires payment of the statutory household premium, a share Q_{below} of the age-eligible population is insured, and the corresponding pool incurs an average cost of AC_{below} per enrollee. Above the threshold, where coverage is free and individual, the insured share is Q_{above}, with an average pool cost of AC_{above}. The cost of the marginal enrollees can therefore be recovered from these average costs and the associated changes in enrollment.

Specifically, applying equation (2) at the two coverage rates and taking their difference yields the total cost of the individuals enrolled between the two coverage rates:

$$AC_{\text{above}}\, Q_{\text{above}} - AC_{\text{below}}\, Q_{\text{below}} \;=\; \int_{Q_{\text{below}}}^{Q_{\text{above}}} MC(q)\,\mathrm{d}q, \tag{3}$$

where the left-hand side represents the difference in total insured costs across the two coverage rates. Dividing this expression by the mass of individuals whose enrollment is induced by the

14. Fischer, Frölich, and Landmann (2023) instead construct an expected-cost index from detailed baseline covariates collected before enrolment. They do so for two reasons: their claims data contain only 39 claims, making claim-based cost curves too imprecise, and an index based on baseline characteristics cannot incorporate any utilisation response to coverage. Their index is estimated from 334 inpatient events reported in a high-frequency phone survey. Neither consideration transfers cleanly to this setting. Precision is not the binding constraint here, since the claims data contain 32.9 million reimbursements. At the same time, the registration records contain no comparable baseline health information, and individuals induced to enrol by the waiver enter the administrative data only once they become insured, leaving no pre-enrolment history from which an analogous index could be constructed. Realised claims are therefore both the feasible cost measure in this setting and the convention adopted by Finkelstein, Hendren, and Shepard (2019). The trade-off is that any contamination from changes in utilisation, which their index design rules out, must instead be addressed by the sign and institutional argument in Assumption 4 and Section 5.4.

waiver, $\Delta Q = Q_{\text{above}} - Q_{\text{below}}$, yields the average cost of these marginal enrollees:

$$\overline{MC} \;\equiv\; \frac{1}{\Delta Q} \int_{Q_{\text{below}}}^{Q_{\text{above}}} MC(q)\,\mathrm{d}q \;=\; \frac{AC_{\text{above}}\,Q_{\text{above}} - AC_{\text{below}}\,Q_{\text{below}}}{\Delta Q}. \tag{4}$$

All terms on the right-hand side are identified from the research design. Thus together with the contract-equivalence condition examined in Section 5.4, $\overline{MC}$ identifies the average cost of the enrollees whose insured status changes at the threshold. Importantly, the object identified by the design is an average rather than the marginal cost schedule itself. Whereas $MC(\cdot)$ is a function describing costs across the enrollment distribution, $\overline{MC}$ is a scalar corresponding to the average value of this function over the interval of enrollment induced by the waiver. Consequently, the design identifies the latter object but not the former, and the slope of $MC(\cdot)$ is therefore not identified.

The two average-cost measures nevertheless suffice to characterize the relative cost of the marginal and inframarginal enrollees. Because the insured pool above the threshold consists of the pool below the threshold together with the additional enrollees induced by the waiver, AC_{above} can be expressed as a weighted average of AC_{below} and $\overline{MC}$. It must therefore lie between these two quantities. Consequently, average cost decreases at the threshold if and only if

$$\overline{MC} \;<\; AC_{\text{below}}, \tag{5}$$

so that a reduction in average cost is equivalent to the marginal enrollees induced by the waiver having lower costs than the enrollees retained under the paid regime. This is the comparison reported by Finkelstein, Hendren, and Shepard (2019) at each of their premium discontinuities, and it is the sense in which I use the term adverse selection throughout the analysis.

5.3. Results

The enrollees induced by the waiver have an estimated cost of NPR 3,358 per enrollee, compared with NPR 5,269 among the pool already enrolled under the paid regime. The resulting ratio is 0.637, corresponding to a difference of NPR 1,911 per enrollee. The closest comparable estimate is 0.59, obtained at a premium discontinuity in the Massachusetts subsidized exchange (Finkelstein, Hendren, and Shepard 2019). [15]

The estimate indicates substantial adverse selection into insurance among the enrollees in the paid pool: $\overline{MC}$ lies below AC_{below} in all 500 bootstrap draws. The underlying inputs are reported in Table 6. Coverage increases from 0.219 of the eligible population immediately below the threshold to 0.423 immediately above it, implying that the waiver induces enrollment of $\Delta Q = 0.204$, or approximately one-fifth of the eligible population. Over the same interval, average cost among the insured falls from NPR 5,269 to NPR 4,348. The resulting decline of NPR 922 is generated by the same local-linear limits of realized claims that identify the claims discontinuity reported in Table 5. Thus, the two exhibits should be interpreted as different representations of the same local estimate rather than as independent pieces of evidence.

Table 6 — Coverage and costs at the age-70 threshold

	Point	Boot. SE	95% CI
Panel A: Coverage			
Q_{below}	0.219	0.001	[0.217, 0.222]
Q_{above}	0.423	0.001	[0.420, 0.426]
ΔQ (induced margin)	0.204	0.002	[0.200, 0.208]
Panel B: Costs (NPR per enrollee)			
AC_{below}	5,269	106	[5,077, 5,477]
AC_{above}	4,348	65	[4,237, 4,491]
$AC_{\text{above}} - AC_{\text{below}}$	-922	122	[-1,132, -658]
$\overline{MC}$ (induced enrollees)	3,358	175	[3,050, 3,739]
$AC_{\text{below}} - \overline{MC}$	1,911	255	[1,363, 2,366]
$\overline{MC}/AC_{\text{below}}$	0.637	0.043	[0.563, 0.729]

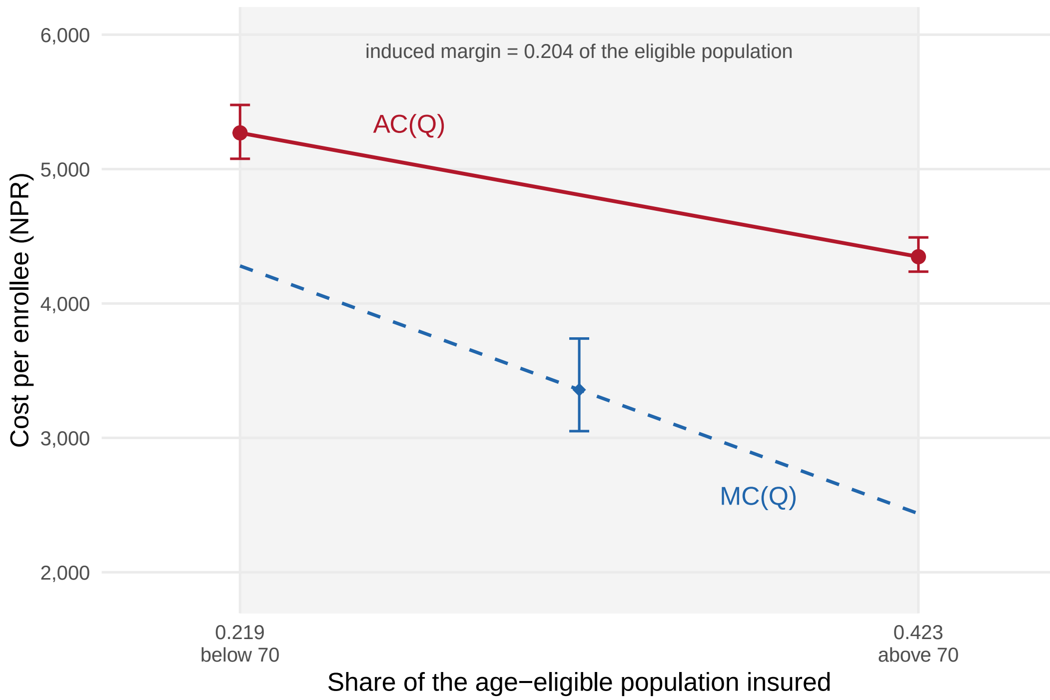
Notes: Coverage is the point-in-time insured stock of Section 4.1. Costs are realised claims per enrolment spell, taken as local-linear limits on age in days at $h = 485$ days, so $AC_{\text{above}} - AC_{\text{below}}$ is the discontinuity in total claimed reported in Table 5. $\overline{MC}$ is the average cost of the enrollees the waiver draws in, recovered from the identity in the text. Households are resampled with replacement, $B = 500$ times. Each replication uses the same resampled households for the claims and for the insured counts, so the average costs and the coverage shares move together rather than independently. Population denominators and the mortality correction are administrative inputs and are held fixed across draws. Intervals are bias-corrected percentile intervals, and the interval for a ratio is computed from its own bootstrap distribution. $\overline{MC}$ falls below AC_{below} in 100.0 percent of draws. Sensitivity to the bandwidth used for the average costs is in Table C1.

Figure 6 plots the two average-cost points, the interval in coverage induced by the waiver,

15. Their design does not report this ratio directly; I construct it from Figures 7B and 8B for fiscal year 2011. At the 150-percent-of-poverty threshold, enrollment in the composite high-coverage plan falls from 80 to 64 percent as the premium rises, while average cost among those who remain rises from $361 to $393 per month.

and $\overline{MC}$ at the midpoint of that interval. The dashed line shows the marginal cost implied by assuming that average cost is linear in coverage. It is included for orientation only: its value at the midpoint coincides with $\overline{MC}$ by construction, not because the slope of marginal cost is identified.

Figure 6 — Average and marginal cost over the coverage induced at age 70



Note: Cost per enrollee against the share of the age-eligible population insured. The two points are local-linear limits of realised claims per enrolment spell on either side of age 70, at $h = 485$ days, plotted at the coverage rates in Section 4.1. Bars are 95 percent bias-corrected bootstrap intervals. The diamond is $\overline{MC}$ from equation (4), plotted at the midpoint of the induced coverage interval. The dashed line is the marginal cost implied by a linear average-cost schedule; its slope is not identified, and it coincides with $\overline{MC}$ at the midpoint by construction (Fischer, Frölich, and Landmann 2023). Shading denotes the induced interval.

The ratio is sensitive to bandwidth. Across bandwidths of 365 to 1,095 days (Table C1), $\overline{MC}$ ranges from NPR 3,726 to NPR 2,924, while the ratio ranges from 0.728 to 0.554. The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth. The magnitude of the gap nevertheless varies with bandwidth, as is typical for regression discontinuity estimates, so I interpret the ratio as an estimate with an interval rather than as a fixed parameter of the programme.

The four inputs to equation (4) come from two designs. Average costs are local-linear limits of realised claims per enrolment spell, while coverage shares are point-in-time stocks from Section 4.1. The two measures differ in population, running variable, bandwidth, and treatment of decedents, so $AC \cdot Q$ should not be interpreted as the total cost of a single enumerated pool. A common denominator error, however, cancels: scaling both Q_{below} and Q_{above} by the same factor leaves $\overline{MC}$ and the ratio unchanged. Only errors that differ across the threshold can therefore affect the estimates.

These estimates are local. The compliers are individuals near age 70, and the cost-participation relationship identified here pertains to this group. Extrapolating the selection pattern to other ages, or to never-takers who decline free coverage, requires assumptions that the design does not test.

5.4. *Threats to Interpretation*

Assumption 2 is the standard regression discontinuity requirement and is defended by the broader design in Section 6.3. Assumption 3 is a measurement convention. The substantive identifying burden therefore rests on Assumptions 1 and 4, the contract-equivalence condition, and the household as the relevant enrollment decision unit.

Continuity of the insured pool (Assumption 1). Free coverage weakly dominates paid coverage in both the premium and benefit ceiling, so an individual insured at 69 should not prefer to become uninsured at 70. The relevant assumption, however, concerns the enrollment process rather than preferences: individuals insured just below the threshold must remain in the insured pool just above it. This requires the conversion from a listed household member to an individual policyholder to occur without systematic failure. If conversion failed disproportionately among frail, high-cost payers, their exit at 70 could mechanically generate the observed cost decline. Two pieces of evidence argue against such a pattern. First, coverage nearly doubles at the threshold, from 0.219 to 0.423 of the eligible population, which is difficult to reconcile with widespread attrition during conversion. Second, among enrollees whose coverage lapses and later resumes, those whose gap spans age 70 have NPR 8,757 in claims before 70, compared with NPR 19,354 for others (Section 6.3). Thus, enrollees who leave the pool around the threshold are less than half as costly as those who do not, the opposite of what would generate the estimated selection pattern. Neither statistic directly measures one-way exit at 70, so together they constrain rather than eliminate this concern.

The contract change at 70 (contract equivalence). The benefit ceiling changes at 70 from a household total of NPR 100,000 to an individual entitlement, so the contracts are not literally identical across the threshold. If the ceiling were binding, the more generous contract above 70 would mechanically raise AC_{above}, working against the observed decline. The ceiling is not binding in practice. Because the pre-70 ceiling is shared across household members, the relevant test uses household totals: 0.48 percent of household-spells reach 80 percent of the ceiling, compared with 0.17 percent above the threshold (Table C2). The constraint is therefore close to irrelevant on both sides, implying that the decline in average cost is not driven by the relaxation of the ceiling. Consistent with weak enforcement, the largest individual spell below the threshold records NPR 158,117 in claims, well above the nominal ceiling.

Behavioural response (Assumption 4). Because $\overline{MC}$ is based on realised claims, it could reflect changes in healthcare use induced by the waiver rather than differences in underlying health. The standard price-of-care channel is absent: care is cashless at the point of service on both sides of the threshold, and copayments were introduced only in 2025, after the sample period.[16] Two channels could nevertheless increase utilisation above the threshold. The waived premium relaxes the household budget, while the individual entitlement removes the shadow price associated with drawing down an allowance shared with relatives. Both effects increase utilisation and therefore raise AC_{above} and $\overline{MC}$ through equation (4). The estimated marginal cost is consequently an upper bound on the cost of the marginal enrollees absent behavioural responses, making the finding that $\overline{MC} < AC_{\text{below}}$ conservative. Only a response that reduces utilisation above the threshold could overturn the result, and it would need to be large. Let τ_M denote the change in costs, at the threshold, for individuals who would enroll under either price, and let c_E denote the cost of entrants. Equation (4) then gives $\overline{MC} = c_E + (Q_{\text{below}}/\Delta Q)\,\tau_M$. Since $Q_{\text{below}}/\Delta Q = 1.07$, raising c_E above AC_{below} would require $\tau_M \approx -1{,}780$: enrollees just below the threshold would have to spend roughly one-third more than AC_{below} purely because they paid a premium.[17]

The household as the decision unit (single ordering). Below the threshold, the premium is a

16. Statutory basis and dates are reported in Section 2.

17. This is not a stronger identifying requirement than in Finkelstein, Hendren, and Shepard (2019). Their selection contrast holds the plan fixed across each premium discontinuity and therefore does not require an argument about behavioural responses. Their discussion of moral hazard instead addresses willingness to pay relative to cost. In my setting, the threshold also changes the terms of coverage, which is why Assumption 4 and the ceiling evidence are relevant.

household bundle: NPR 3,500 covers up to five members, so adding a member to a household of four has no marginal premium, while an unattached individual faces the full NPR 3,500. There is therefore no common individual price, and the insured population below the threshold need not correspond to the highest segment of an individual willingness-to-pay distribution. The observed selection pattern can instead arise if households are more likely to enroll when they contain a high-cost elder, loading the paid pool with high-cost members through the household decision alone. Household composition does change at the threshold, with mean enrolling-household size falling sharply once coverage becomes free and individual. The estimates should therefore be interpreted as comparing the marginal and inframarginal groups defined by the waiver, rather than as identifying two points on an individual cost curve.

6 — Robustness and Threats

This section presents robustness and falsification tests for the main estimate. I first show that the discontinuity is policy-induced: it is absent in all cohorts that enrolled before the free-premium provision was introduced. I then examine the main choices inherent to the local-linear design, showing that the results are stable across bandwidths, donut specifications, kernel weights, and polynomial orders, and that inference does not depend on the clustering assumption. The remaining subsections address the principal threats to interpretation, including manipulation of the running variable, and strategic timing of re-entry.

6.1. Falsification: Pre-Policy Placebo

The discontinuity is absent from cohorts that enrolled before the free-premium provision was introduced. NHIP waived the premium at age 70 beginning 14 April 2019. Splitting the sample by enrolment quarter and re-estimating the RD therefore provides eight pre-policy placebo estimates and an event study over the twenty subsequent quarters. Figure 7 reports the results for all four claims margins.

Before the waiver, claims show no systematic discontinuity at age 70. The total-claims estimate straddles zero across all eight pre-waiver quarters and is never statistically distinguishable from zero; the other three margins are similarly flat. Three of the 32 pre-waiver estimates are significant at the 5 percent level, consistent with what would arise from 32 independent null hypotheses, and two of these three have the opposite sign from the policy prediction.[18]

The discontinuity emerges in the first treated quarter and persists thereafter. Enrollees starting in May 2019, the first cohort facing a zero premium at age 70, already exhibit substantially lower claims above than below the cutoff. All twenty post-policy quarters have negative and statistically significant estimates at the 1 percent level for total claims, with the other three claims margins switching at the same date. Thus, both the coverage and claims responses begin when the waiver takes effect.

The figure uses a fixed 1,095-day window across quarters and outcomes, which is wider than the MSE-optimal bandwidth used in the main specification. A common window ensures that the quarterly estimates represent comparable local contrasts rather than different age ranges selected quarter by quarter. It also improves the power of the placebo by reducing the average pre-policy standard error for total claims. Because the window is wider, the quarterly estimates are larger in magnitude than those reported in Table 5; the figure is intended to establish the timing of the discontinuity, rather than its preferred magnitude.

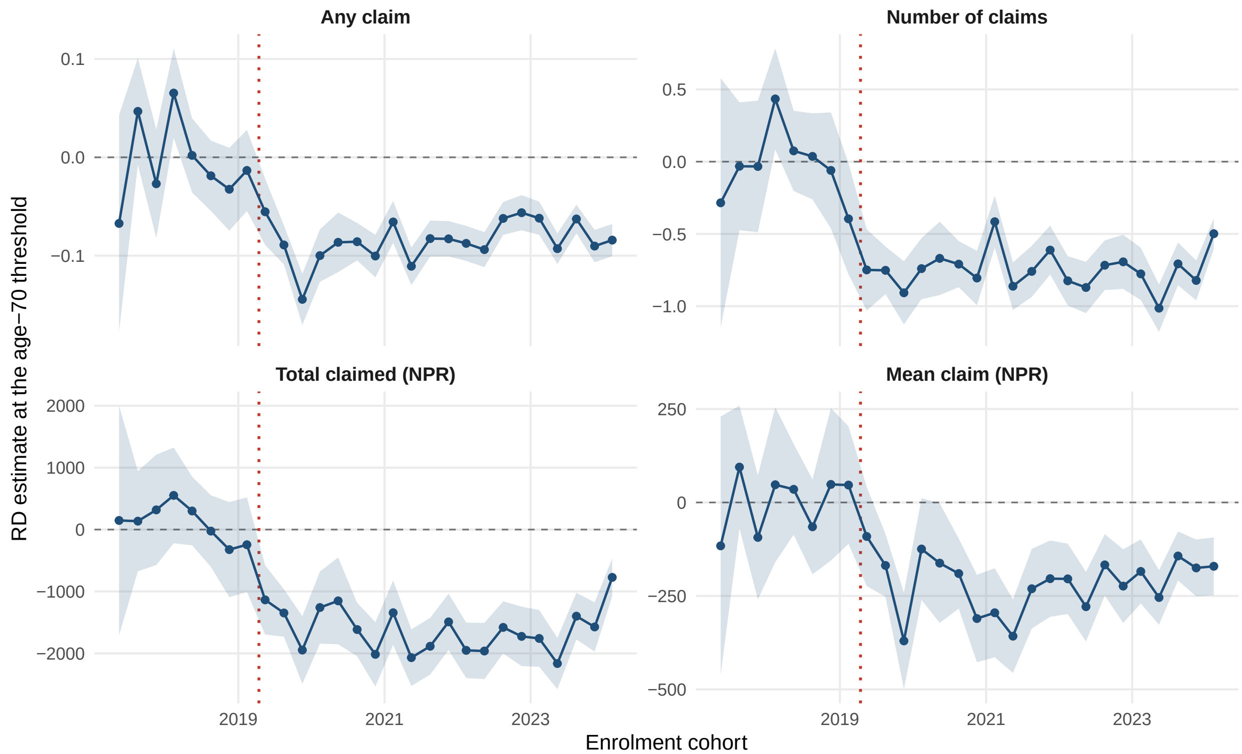
6.2. *Specification Robustness*

The estimate is robust to the main specification choices in the local-linear design. Table D1 in Appendix D.1 re-estimates all four claims outcomes across bandwidths from 180 to 1,095 days, two donut specifications, a triangular kernel, and a quadratic polynomial. Panel E reports the main specification, using the MSE-optimal bandwidth selected by Calonico, Cattaneo, and Titiunik (2014) for each outcome. Across the 96 estimates, 92 are negative and 83 are significant at the 5 percent level.

The donut specifications strengthen rather than attenuate the discontinuity. Dropping the quarter immediately below the cutoff, which removes observations potentially affected by short-run timing around the threshold, increases the discontinuity in both total claimed and any claim

18. The third is the number-of-claims estimate for February 2019, the final quarter before the waiver. Its magnitude is smaller than the estimate in the following quarter, which persists thereafter, while spending is flat in that quarter. Because I cannot separately date the waiver's announcement and implementation, this point could reflect anticipation or noise.

Figure 7 — The discontinuity appears only after the 2019 policy onset



Note: RD estimate at the age-70 threshold by enrolment quarter, with 95 percent confidence intervals. Local linear with separate slopes on each side, uniform kernel, no covariates, clustered on the household, estimated within a fixed window of $h = 1{,}095$ days for every quarter and every panel. The dotted line marks April 2019, when the free-at-70 provision took effect; the first cohort to face it starts in May 2019. Enrolment cohorts run from the 2017 cycle to February 2024. The final point is the smallest of the treated estimates, and that cohort's insurance year is only partly covered by the reimbursement records, which compresses claims on both sides of the cutoff. The estimates and standard errors are reported in Table B1.

relative to the corresponding specification without the donut. Dropping the quarter on both sides produces the same pattern. If manipulation at the threshold were generating the result, removing observations nearest the cutoff should attenuate the estimate; instead, those observations appear to attenuate it.

At bandwidths wider than the MSE-optimal window, the estimated magnitudes increase. These wider windows incorporate more of the age gradient in claims and therefore introduce greater potential specification bias rather than simply adding information. The triangular-kernel estimates, which place greater weight on observations nearest the cutoff, remain negative and significant at the same window.

The 13 insignificant estimates are concentrated in specifications with limited identifying power.

Eleven are donut specifications at the two narrowest bandwidths, where removing 91 days on one or both sides leaves little variation with which to estimate the slope. At $h = 180$, the symmetric donut creates a 182-day gap, and the standard error for total claimed is about four times that of the specification without the donut. All four positive estimates occur in this same corner. The remaining two insignificant estimates use a quadratic at $h = 485$, where the second-order term is estimated over a relatively narrow window and is consequently imprecise; at wider bandwidths, the quadratic estimates are negative and significant. No specification with sufficient precision to detect the effect reverses its sign. Appendix D.2 reports these specification with standard errors clustered on the value of the running variable and with permutation-based inference.

6.3. Manipulation and Strategic Timing

The running variable is date of birth, recorded in administrative registration records and determined by aging rather than choice. It is anchored to the citizenship certificate or national identity document required for NHIP enrollment, which is issued long before the waiver becomes relevant. Manipulation would therefore require falsifying an identity record rather than timing an enrollment decision. Although misreporting in the registry cannot be ruled out a priori, the density test provides a direct diagnostic.

The standard concern is bunching of the running variable at the cutoff. In this setting, the relevant density is that of the eligible population, not enrollees. An enrollee-level density test would be inappropriate because enrollment responds to the price: the jump in enrollee density at age 70 is precisely the treatment response of interest. Using the census population density and correcting for age heaping, the log-density discontinuity at 70 is -0.046 (SE 0.145) (Cattaneo, Jansson, and Ma 2020), smaller than the corresponding estimates at placebo ages on either side. Appendix A.3 reports both diagnostics. A further check comes from the robustness results in Section 6.2: dropping the quarter on either side of the cutoff, where individuals with falsified dates of birth would be expected to appear, increases rather than attenuates the estimated discontinuity.

A related concern is strategic re-entry. If enrollees allowed coverage to lapse and timed re-enrollment to coincide with the waiver, the pool just above 70 could be selected on anticipated healthcare needs. The enrollment-gap distribution provides little evidence of such behavior. Gaps around age 70 are no longer than elsewhere, and enrollees whose gap spans their seventieth birthday have less than half the pre-70 claims of other enrollees, whereas strategic timing would predict higher anticipated need. Appendix D.3 reports the gap distributions.

7 — Conclusion

National health insurance in low- and middle-income countries relies on enrollment that is voluntary in law or in practice, making the premium a key determinant of who is insured (WHO and Bank 2023; Banerjee et al. 2021). Nepal sets that price to zero at age 70, providing a sharp test of what the premium had been doing. Coverage nearly doubles at the threshold, while claims per enrollee fall across every margin. Because enrollment responds to price, the discontinuity compares two different pools rather than the same pool before and after the threshold. The decline therefore reflects the cost of the pool the programme insures, not a change in healthcare use for a fixed population. The enrollees induced by the waiver cost NPR 3,358 each in paid claims, compared with NPR 5,269 for those already insured under the paid regime, yielding a ratio of 0.637 with a bootstrap interval of [0.563, 0.729]. The premium had therefore screened in higher-cost enrollees below the threshold: the paid programme exhibits adverse selection. This direction is consistent with the experimental evidence from a comparable setting, where time-limited subsidies in Indonesia also attracted enrollees with lower expected costs (Banerjee et al. 2021).

The waiver expands coverage precisely where access gaps are larger. The enrollees above the threshold live 0.54 km farther from their assigned first point of contact, have 0.74 fewer empanelled facilities within 10 km of their ward, and live in local units with poverty headcounts 1.7 percentage points higher. Thus, the additional coverage reaches populations that appear more geographically and economically underserved, while costing less per enrollee once insured.

The threshold is particularly informative because it operates within the programme's existing institutional safeguards. NHIP bundles coverage at the household level, limiting selection within households (Fischer, Frölich, and Landmann 2023), and restricts enrollment to fixed quarterly windows. Both features are already in place below age 70, yet the insured pool remains adversely selected. The waiver therefore does not create the selection problem; it reveals the selection already generated by the premium. The enrollees attracted by a zero price are precisely those whom the positive price had excluded. In Indonesia, the binding constraint was administrative capacity, with more than half of attempted enrollments failing to complete (Banerjee et al. 2021); in Nepal, the pool is adversely selected regardless of the safeguard, leaving the price as the relevant margin. A programme can therefore be administratively well run and still use a price that determines who enters its risk pool.

The fiscal implication is clearer than the welfare implication. For roughly every enrollee the waiver adds, one was already insured under the paid regime and now pays nothing. Approximately half of the fiscal cost therefore subsidizes existing coverage, while the remainder finances newly insured enrollees whose claims cost NPR 3,358 per year. Valuing the welfare gains from this expansion would require a measure of willingness to pay for coverage below age 70, but the household premium schedule provides no single individual price (Section 5.4). The discontinuity therefore identifies the cost of expanding coverage without identifying its welfare value.

The composition of the insured pool improves as coverage expands: the lower price attracts enrollees whose costs are below those of the existing pool. The fiscal position, however, deteriorates. A waiver lowers average cost per enrollee while increasing the total expenditure the programme must finance, precisely as premium revenue ceases to increase. The binding constraint therefore shifts from the composition of the pool to the resources available to finance it. Nepal appears to have reached this constraint. Arrears owed to empanelled hospitals exceeded NPR 16 billion by early 2026, and more than 50 hospitals had suspended acceptance of insurance cards by early 2026, while the Board responded by capping outpatient coverage rather than expanding the contribution base (The Kathmandu Post 2026a, 2026b). A programme can extend coverage to people whom a positive price excludes only by removing that price, but sustaining such expansion requires a pool that also contains members willing or required to pay. Where enforcement is feasible, compulsory enrollment—such as through formal employment—is the one instrument that can expand coverage and revenue simultaneously. This study does not test that alternative, but it shows what other margins leave unchanged: restricting enrollment timing or requiring enrollment at a particular unit does not alter the underlying composition of the pool and is adversely selected regardless.

Three limitations qualify these conclusions. First, the estimates are local to age 70 and to the enrollees induced by a zero price. Second, they identify enrollment and claims costs, leaving any moral-hazard response among marginal enrollees unmeasured. Third, the magnitude of the cost gap varies with bandwidth, with the ratio ranging from 0.554 to 0.728 across one- to three-year windows, while reimbursement records understate costs on both sides as claims accrue. The ordering that defines adverse selection survives these concerns, which motivates reporting the ratio rather than treating its level as a structural parameter. The claims register cannot determine whether the lower cost of marginal enrollees reflects better health or lower healthcare use, nor whether the waiver improves the health or financial protection outcomes that insurance is intended to provide. These limitations do not alter the central result of the age-70 discontinuity: it identifies what the government buys, and for whom, when it makes insurance free.

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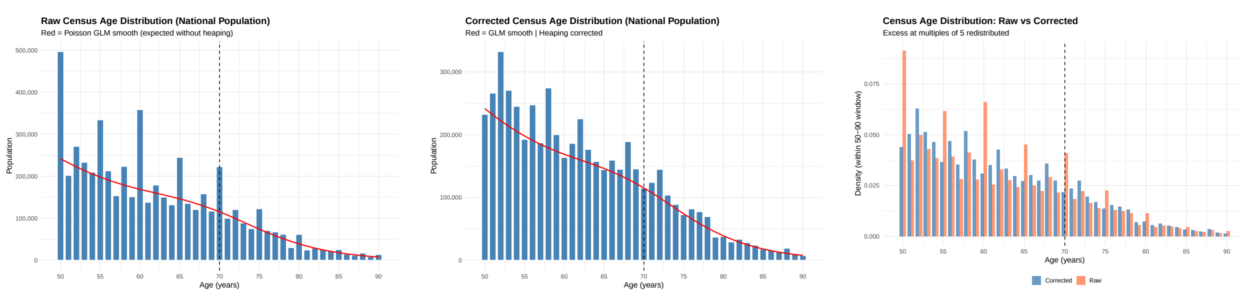
Appendix A — Design Validity

A.1. Census Age-Heaping Correction

The covariate-balance and density tests in Section 3 and Appendix A.3 use the 2021 Nepal Census, where respondents disproportionately report ages in multiples of five. Because age 70 is both the policy threshold and a heaping point, raw counts overstate the population at 70 and understate counts at adjacent ages. I correct this using stochastic redistribution (Heitjan and Rubin 1990). A quasi-Poisson model fitted to single-year counts at non-heaped ages predicts a smooth age density through the heaping points. The excess mass at each heaped age—the difference between observed and predicted counts—is then redistributed across neighboring ages in proportion to the predicted density. Figure A1 compares the raw and corrected counts.

The procedure is repeated $M = 50$ times with independent redistributions. Estimates using the corrected counts are computed separately within each draw and combined using Rubin's rules (Rubin 1987), so the reported standard errors incorporate both sampling uncertainty and uncertainty from the age-heaping correction.

Figure A1 — Age heaping in the 2021 census and its correction



Note: Single-year population counts from the 2021 Nepal Census. Left: raw counts, with spikes at multiples of five. Centre: counts after stochastic redistribution. Right: the two overlaid. The age-70 spike is comparable to spikes at non-policy multiples of five.

The covariate-balance results are robust to the choice of cutoff, the kernel and the polynomial order. The tables below re-estimate balance using placebo cutoffs, including age 63 (A4) and 69 (A3). The age-70 estimates remain statistically indistinguishable from zero throughout, apart from literacy and wealth quintiles.

Table A1— Covariate Balance at Age 70: Sensitivity to Kernel and Polynomial Order

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-70	Uniform	Triang.	Uniform	Triang.
Female	0.509	0.013*	0.012	0.011	0.034*
		(0.007)	(0.009)	(0.015)	(0.020)
High Caste	0.478	0.000	−0.007	−0.016	0.008
		(0.007)	(0.009)	(0.015)	(0.021)
Hindu	0.810	0.010*	0.008	0.006	0.036**
		(0.006)	(0.007)	(0.011)	(0.016)
Ever married	0.988	−0.000	0.001	0.002	0.003
		(0.002)	(0.002)	(0.003)	(0.005)
Has disability	0.064	0.000	−0.003	−0.008	−0.009
		(0.003)	(0.004)	(0.007)	(0.010)
Household size	5.204	0.096	0.168	0.243	0.491
		(0.113)	(0.126)	(0.208)	(0.372)
Economically inactive (aged)	0.354	0.009	−0.000	−0.012	0.000
		(0.007)	(0.009)	(0.014)	(0.020)
Literate	0.339	−0.013*	−0.025***	−0.036***	−0.112***
		(0.007)	(0.009)	(0.014)	(0.019)
Wealth quintile	2.875	−0.039*	−0.055*	−0.076*	−0.267***
		(0.022)	(0.029)	(0.044)	(0.061)
Bandwidth (years)			4		
Observations			138,314		

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (`feols`) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 70. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin's combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2 — Joint covariate balance at the age-70 threshold

Joint Wald statistic (D_1)	0.654
Reference distribution	$F(9, 9941)$
p-value (joint)	0.751
Bonferroni-adjusted minimum individual p	1.000
Covariates tested jointly	9
Redistribution draws (M)	50

Notes: All nine covariates from the balance table are estimated in a single stacked local linear regression (uniform kernel, MSE-optimal bandwidth, census weights, SEs clustered by household across equations), and the nine treatment coefficients are tested jointly. Estimates are combined across M stochastic age-redistribution draws with the multivariate Rubin rules (D_1 statistic; Li, Raghunathan and Rubin 1991). Covariates are standardized within the window.

A.2. Covariate Balance: Age Profiles

Figure A2 displays the age profile of each census covariate, with separate linear fits on each side of the threshold. The visual continuity at age 70 matches the small, jointly insignificant point estimates of Section 3.2.

A.3. Density of the Eligible Population

The standard manipulation concern in a regression discontinuity design is bunching of the running variable at the cutoff. Here, the relevant object is the density of the *eligible population*—everyone aging through 70—rather than the density of enrollees. An enrollee-level McCrary-style test is inappropriate because enrollment itself responds to the price change: the jump in enrollee density at 70 is the treatment response of interest, not evidence of manipulation (McCrary 2008). The running variable is date of birth recorded in administrative registration records and determined by aging rather than choice. It is anchored to the citizenship certificate or national identity document required for NHIP enrollment, although misreporting in the registry cannot be ruled

Table A3 — **Placebo Balance Test at Age 69 (No Policy Change)**

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-69	Uniform	Triang.	Uniform	Triang.
Female	0.510	0.003	-0.001	-0.008	-0.024
		(0.007)	(0.008)	(0.013)	(0.020)
High Caste	0.479	0.006	0.001	-0.010	-0.031
		(0.007)	(0.008)	(0.013)	(0.020)
Hindu	0.813	0.011^{**}	-0.003	-0.023^{**}	-0.037^{**}
		(0.005)	(0.006)	(0.010)	(0.015)
Ever married	0.988	0.001	-0.001	-0.004	-0.002
		(0.001)	(0.002)	(0.003)	(0.005)
Has disability	0.061	0.001	0.004	0.009	0.001
		(0.003)	(0.004)	(0.006)	(0.009)
Household size	5.313	0.165^{*}	-0.091	-0.479	-0.281
		(0.090)	(0.143)	(0.335)	(0.443)
Economically inactive (aged)	0.328	0.042^{***}	0.013^{*}	-0.030^{**}	-0.014
		(0.006)	(0.008)	(0.012)	(0.019)
Literate	0.339	-0.036^{***}	0.019^{**}	0.106^{***}	0.089^{***}
		(0.006)	(0.008)	(0.013)	(0.019)
Wealth quintile	2.867	-0.077^{***}	0.054^{**}	0.248^{***}	0.242^{***}
		(0.020)	(0.023)	(0.039)	(0.060)
Bandwidth (years)			4		
Observations			144,613		

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (`feols`) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 69. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin's combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4— Placebo Balance Test at Age 63 (No Policy Change)

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-63	Uniform	Triang.	Uniform	Triang.
Female	0.508	0.009	0.003	-0.005	-0.052^{***}
		(0.006)	(0.008)	(0.013)	(0.019)
High Caste	0.469	0.006	-0.002	-0.010	-0.050^{***}
		(0.006)	(0.007)	(0.012)	(0.019)
Hindu	0.812	-0.000	-0.005	-0.009	-0.050^{***}
		(0.005)	(0.006)	(0.010)	(0.015)
Ever married	0.986	-0.002	-0.002	-0.002	-0.007
		(0.001)	(0.002)	(0.003)	(0.004)
Has disability	0.051	0.001	-0.001	-0.003	-0.002
		(0.003)	(0.003)	(0.005)	(0.008)
Household size	5.307	-0.072	-0.086	-0.041	-0.473
		(0.105)	(0.127)	(0.205)	(0.364)
Economically inactive (aged)	0.133	-0.014^{***}	0.031^{***}	0.103^{***}	-0.023
		(0.004)	(0.006)	(0.009)	(0.014)
Literate	0.400	-0.002	-0.017^{**}	-0.049^{***}	0.146^{***}
		(0.006)	(0.007)	(0.012)	(0.019)
Wealth quintile	2.925	0.018	-0.010	-0.069^{*}	0.395^{***}
		(0.018)	(0.023)	(0.037)	(0.057)
Bandwidth (years)			4		
Observations			174,496		

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (`feols`) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 63. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin's combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

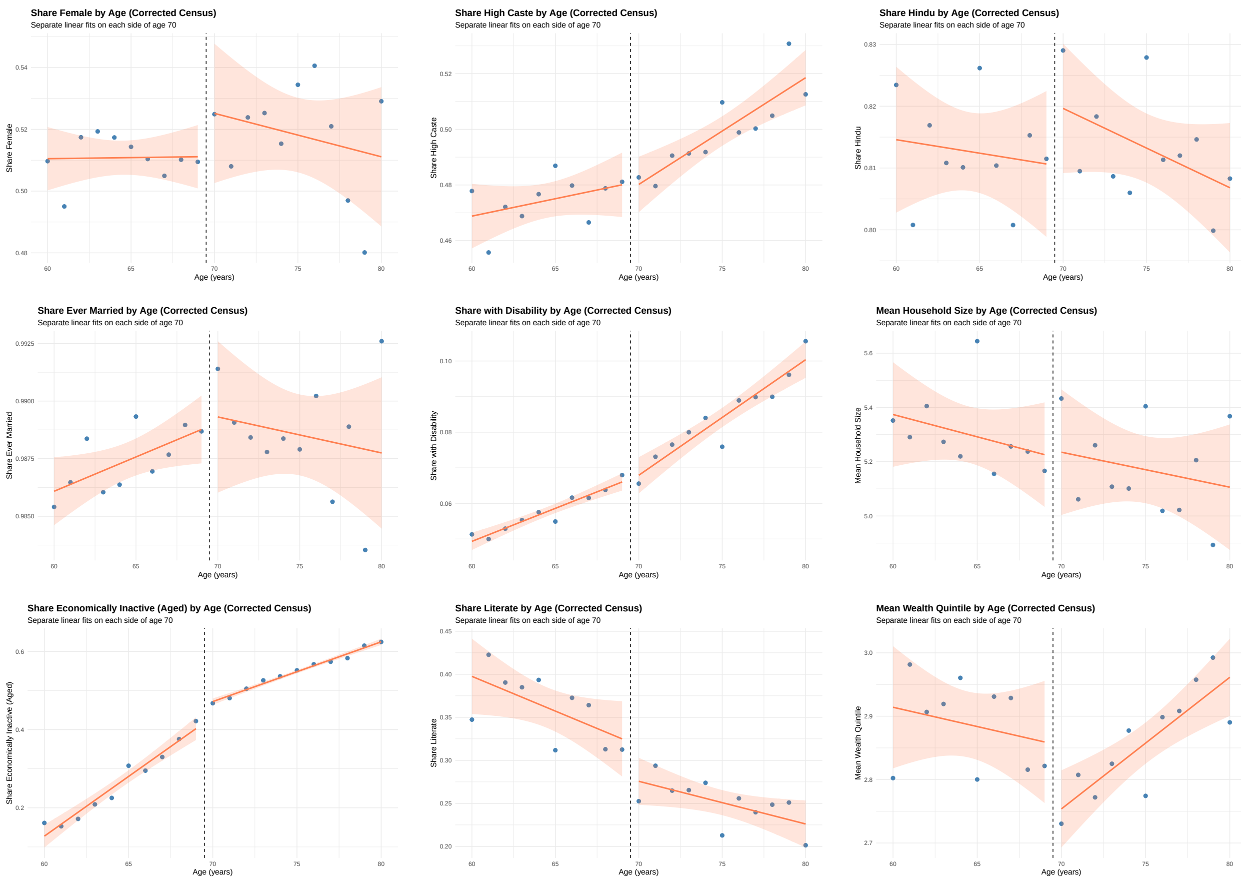


Figure A2 — Covariate age profiles around the age-70 threshold (2021 Nepal Census). The panels are the nine covariates of Table 3, in the order they appear there. Each shows the weighted mean by age with separate linear fits on each side of the cutoff; ages corrected for heaping (Appendix A.1).

out a priori. The census density test therefore addresses manipulation of the eligible population, while enrollee-side irregularities are assessed through the placebo tests in Section 3.3 and the inference diagnostics in Appendix D.2.

I test the smoothness of the census population density at age 70, correcting for age heaping. Table A5 reports both diagnostics. Panel A compares observed single-year counts with a smooth fit estimated excluding multiples of five. The raw count at age 70 is 1.94 times the smooth prediction, within the 1.42–2.18 range for non-policy multiples of five and below the corresponding ratio at age 60 (2.18), where no insurance policy applies. The spike is therefore consistent with round-age reporting rather than strategic manipulation at the policy threshold. Panel B estimates a quasi-Poisson local-linear model using the heaping-corrected counts over ages 62–78. The estimated log-density discontinuity at 70 is -0.046 (SE 0.145, $p = 0.753$), compared with larger-magnitude estimates at placebo ages 67 and 73 (0.296 and -0.226, respectively) (Cattaneo, Jansson, and Ma 2020). The age-70 estimate is therefore small relative to the variation in corrected population density at ages without policy changes. Figure A3 plots the corrected density.

Table A5 — Population density at the age-70 threshold

Panel A: Raw census — excess mass at heaping points (observed/expected)							
Age	55	60	65	70	75	80	85
Excess ratio	1.74	2.18	1.69	1.94	1.68	1.62	1.42

Panel B: Corrected census — log-density jump (Rubin-pooled over 50 draws)			
	At 70	At 67 (placebo)	At 73 (placebo)
Jump	-0.0456	0.2957	-0.2256
SE	(0.1447)	(0.1514)	(0.1468)

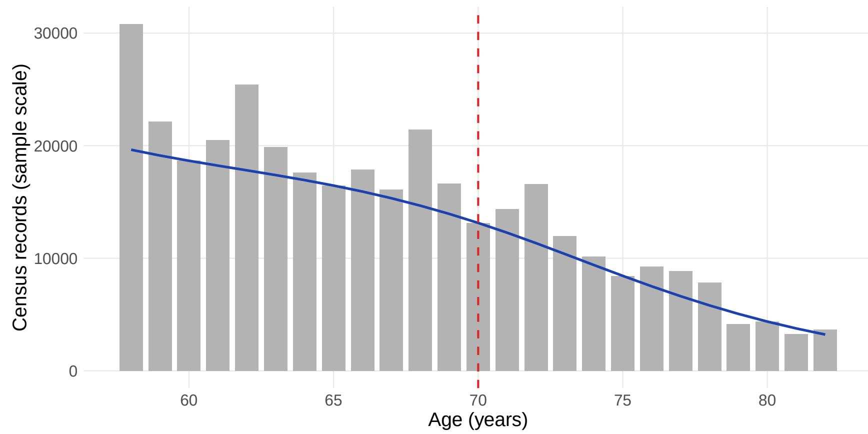
Notes: Panel A compares observed single-year age counts to a quasi-Poisson spline fit estimated excluding multiples of five: the excess at 70 is in line with the excess at 65 and 75, which carry no policy discontinuity, indicating round-age reporting rather than manipulation. Panel B fits a quasi-Poisson local linear model to heaping-corrected single-year counts (ages 62–78) and reports the log jump at the stated cutoff, combined across 50 redistribution draws with Rubin's rules. The enrollee-level density is not tested: its jump at 70 is the enrollment response to the price change, not manipulation of date of birth.

Appendix B — Additional Claims Results

B.1. The Remaining Claims Margins

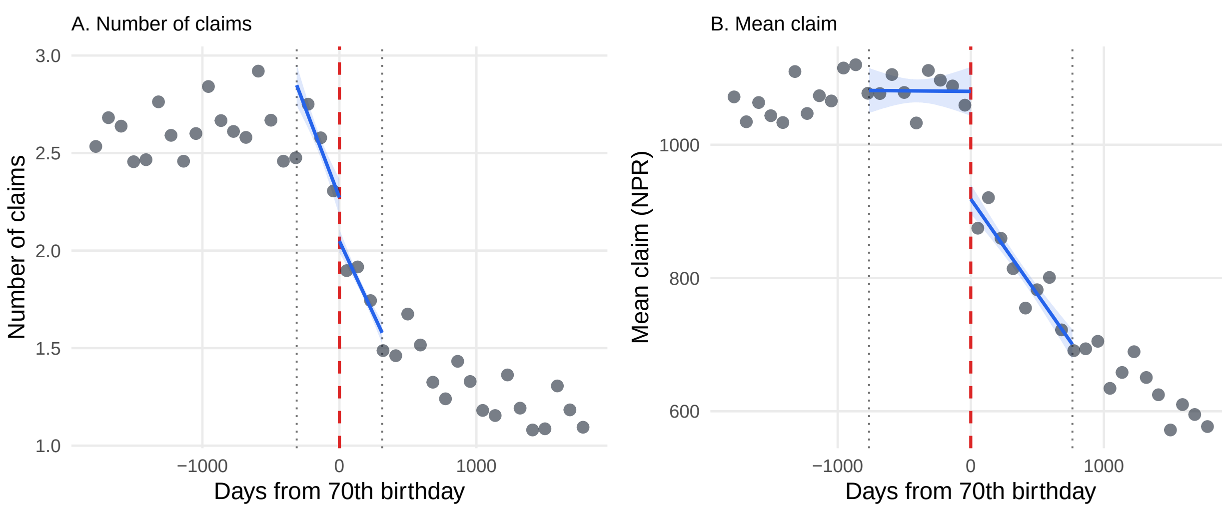
Figure B1 plots the two claims margins the main text leaves out, the number of claims and the mean claim, where Figure 4 shows any claim and total claimed. Both step down at the threshold in the same way, which is the graphical counterpart of Table 5.

Figure A3 — Distribution of Eligible Population



Note: Single-year age counts of the eligible population from the 2021 Nepal Census, corrected for age heaping (Appendix A.1). The dashed vertical line marks age 70. Table A5 reports the formal test of the log-density jump at the threshold and the placebo jumps at ages 67 and 73.

Figure B1 — The discontinuity on the remaining two claims margins



Note: Drawn as Figure 4 is drawn. Binned means in quarterly bins of age at the spell start date, shown over twenty quarters on each side of the seventieth birthday, with local linear fits estimated on each side of the cutoff on the spell-level data inside each panel's own bandwidth, a uniform kernel and no covariates or fixed effects. The bandwidth is 312 days for the number of claims and 764 for the mean claim. The gap drawn at the cutoff is the coefficient reported in Table 5.

B.2. *Pre-Policy Event Study Estimates*

Table B1 reports the quarter-by-quarter estimates plotted in Figure 7 for all four claims margins, allowing the pre-policy nulls to be assessed directly rather than inferred from the confidence intervals in the figure.

B.3. *Timing of the First Claim*

Table B2 reports the three timing outcomes discussed in Section 4.2. The running variable, kernel, clustering, and inference follow Table 5, with each outcome using its own MSE-optimal bandwidth. The resulting bandwidths range from 379 to 803 days, so the three estimates use different windows and samples ranging from 86,908 to 114,357 spells.

The two indicator outcomes are defined for every spell, taking a value of zero for enrollees who never file a claim. They therefore capture both whether a claim occurs and how quickly it occurs. Days to the first claim is defined only among enrollees who file a claim, so its below-cutoff mean of 67.7 days is calculated over a pool whose composition changes at the threshold.

B.4. *Distance to Facilities*

Table B3 reports the total-claims discontinuity by distance from the enrollee's ward centroid to the registered first point of contact. All bands use the same 485-day window as the headline estimate, so the specifications differ only in distance.

The estimated discontinuities are NPR -734.7 (SE 186.7) within 5 km, NPR $-1{,}114.0$ (SE 254.8) at 5–10 km, NPR -644.1 (SE 328.9) at 10–15 km, and NPR -933.9 (SE 319.4) beyond 15 km. The pooled estimate is NPR -918.0 (SE 127.3) across 125,859 spells with geocoded wards and facilities, closely matching the headline estimate in Table 5; the small difference reflects the 1.2 percent of spells without geocodes.

The estimates do not differ significantly across distance bands. The nearest and most remote bands differ by only NPR 199, with a standard error of roughly 370, and the point estimates show no monotonic distance gradient. The largest decline occurs at 5–10 km and the smallest at 10–15 km. Thus, the decline in claims does not appear to be driven by the greater remoteness of

Table B1 — RD estimates by enrolment quarter

Enrolment quarter	Any claim	N claims	Total claimed (NPR)	Mean claim (NPR)
	(1)	(2)	(3)	(4)
Panel A: Before the waiver				
May 2017	-0.067 (0.056)	-0.285 (0.440)	146.9 (945.9)	-116.1 (176.6)
Aug 2017	0.047 (0.028)	-0.032 (0.226)	135.7 (412.7)	94.5 (83.9)
Nov 2017	-0.027 (0.028)	-0.033 (0.232)	317.8 (454.9)	-93.5 (84.7)
Feb 2018	0.065 (0.023)	0.434 (0.179)	551.7 (394.4)	47.4 (105.8)
May 2018	0.002 (0.019)	0.075 (0.142)	300.2 (283.1)	35.1 (62.0)
Aug 2018	-0.019 (0.018)	0.036 (0.152)	-24.8 (295.0)	-65.1 (64.7)
Nov 2018	-0.032 (0.021)	-0.061 (0.205)	-324.0 (391.6)	48.2 (104.0)
Feb 2019	-0.013 (0.021)	-0.396 (0.196)	-245.0 (390.1)	46.5 (80.3)
Panel B: After the waiver				
May 2019	-0.055 (0.017)	-0.749 (0.144)	-1,135.5 (285.2)	-90.9 (67.7)
Aug 2019	-0.089 (0.010)	-0.752 (0.084)	-1,346.9 (195.7)	-168.6 (42.8)
Nov 2019	-0.144 (0.013)	-0.908 (0.111)	-1,946.0 (278.5)	-370.4 (65.8)
Feb 2020	-0.100 (0.014)	-0.740 (0.108)	-1,259.5 (297.5)	-124.6 (69.5)
May 2020	-0.086 (0.015)	-0.669 (0.130)	-1,151.5 (357.7)	-162.3 (81.5)
Aug 2020	-0.086 (0.010)	-0.709 (0.082)	-1,615.0 (220.2)	-190.2 (47.7)
Nov 2020	-0.100 (0.011)	-0.805 (0.096)	-2,015.6 (265.6)	-310.5 (59.7)
Feb 2021	-0.066 (0.011)	-0.416 (0.092)	-1,343.4 (265.3)	-295.1 (60.8)
May 2021	-0.111 (0.010)	-0.862 (0.084)	-2,068.5 (233.3)	-357.7 (50.2)
Aug 2021	-0.083 (0.009)	-0.759 (0.090)	-1,884.7 (233.6)	-230.8 (54.4)
Nov 2021	-0.083 (0.009)	-0.612 (0.086)	-1,490.9 (232.6)	-203.9 (52.1)
Feb 2022	-0.087 (0.009)	-0.825 (0.087)	-1,950.9 (228.9)	-204.4 (48.0)
May 2022	-0.094 (0.009)	-0.871 (0.091)	-1,961.7 (231.6)	-278.8 (47.4)
Aug 2022	-0.062 (0.009)	-0.717 (0.088)	-1,580.2 (215.2)	-167.0 (41.9)
Nov 2022	-0.056 (0.009)	-0.693 (0.096)	-1,725.3 (244.9)	-223.8 (50.2)
Feb 2023	-0.062 (0.009)	-0.777 (0.092)	-1,758.6 (233.9)	-184.6 (43.7)
May 2023	-0.093 (0.008)	-1.014 (0.084)	-2,163.5 (211.3)	-254.3 (37.2)
Aug 2023	-0.063 (0.007)	-0.707 (0.075)	-1,398.9 (192.5)	-143.2 (33.4)
Nov 2023	-0.090 (0.008)	-0.822 (0.071)	-1,573.1 (202.6)	-175.1 (38.8)
Feb 2024	-0.084 (0.008)	-0.499 (0.054)	-772.5 (156.2)	-170.7 (39.4)

Notes: The estimates plotted in Figure 7. Each cell is a local linear RD at the age-70 threshold estimated on that enrolment quarter alone, with separate slopes on each side, a uniform kernel, no covariates, and a window fixed at $h = 1{,}095$ days throughout. Standard errors clustered on the household in parentheses. The free-at-70 provision took effect in April 2019, so the first quarter of Panel B is the first cohort to face a zero premium at 70.

Table B2 — RD estimates at the age-70 threshold, timing of the first claim

	Days to first claim (days) (1)	Claim within 30 days (0/1) (2)	Claim within 90 days (0/1) (3)
RD estimate	7.36^{***}	-0.0492^{***}	-0.0574^{***}
	(1.05)	(0.0050)	(0.0064)
Robust p-value	<0.001	<0.001	<0.001
Mean below cutoff	67.7	0.225	0.369
% change	10.9%	−21.9%	−15.5%
Bandwidth (days)	803	440	379
Observations	86,908	114,357	96,149

Notes: Each observation is an enrolment spell. The running variable, specification, bandwidth rule, clustering and inference are those of Table 5. Column 1 is defined only for spells that claim, so it conditions on a post-treatment outcome and is descriptive. Columns 2 and 3 are defined for every spell, non-claimers included, and are zero for a spell that never claims. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

the marginal pool.

Table B3 — RD estimates at the age-70 threshold, total claimed by distance to the first point of contact

	0–5 km (1)	5–10 km (2)	10–15 km (3)	15 km and over (4)	Pooled (5)
RD estimate	-734.7^{***}	$-1{,}114.0^{***}$	-644.1^{*}	-933.9^{***}	-918.0^{***}
	(186.7)	(254.8)	(328.9)	(319.4)	(127.3)
Mean below cutoff	6,103	4,734	4,399	3,477	5,290
% change	-12.0%	-23.5%	-14.6%	-26.9%	-17.4%
Bandwidth (days)	485	485	485	485	485
Observations	62,755	31,540	15,351	16,213	125,859

Notes: Each observation is an enrolment spell. The outcome is the total amount claimed in NPR over the spell, zero for a spell with no claims. Distance is the Haversine distance in kilometres from the enrollee's ward centroid to the registered first point of contact. Each band is fitted separately, and all are fitted on the same bandwidth, the mean-squared-error-optimal choice for total claimed in Table 5, held fixed so that the bands differ only in distance. Column 5 pools the four bands. Standard errors in parentheses are clustered on the household. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix C — Selection and Cost

C.1. Bandwidth Sensitivity of the Cost Estimates

Table C1 recomputes the two average costs at four bandwidths and recovers $\overline{MC}$ from the identity at each bandwidth. The coverage shares are held fixed at their Table 6 values because they are based on the point-in-time insured stock and do not depend on the claims bandwidth.

AC_{below} is stable across specifications, ranging from 5,117 to 5,279, while $\overline{MC}$ declines from 3,726 to 2,924 as the bandwidth widens. From the identity, $\overline{MC}$ equals AC_{below} plus $Q_{\text{above}}/\Delta Q$ times the difference between the two average costs. Since this multiplier is close to two, larger estimated discontinuities in average cost translate into roughly twice as large changes in $\overline{MC}$. The decline therefore reflects the increasing estimated average-cost gap at wider bandwidths, rather than changes in the cost of the pre-existing pool.

The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth, so the qualitative finding is robust to bandwidth choice. The magnitude is less stable: the ratio ranges from 0.554 to 0.728, comparable to the bootstrap interval at the main bandwidth, [0.563, 0.729]. Thus, bandwidth choice and sampling uncertainty generate similar variation in the ratio, but neither approaches one.

Table C1 — The ratio across bandwidths for the average costs

Specification	AC_{below}	$\overline{MC}$	$\overline{MC}/AC_{\text{below}}$
h = 365 days	5,117	3,726	0.728
h = 485 days (main)	5,269	3,358	0.637
h = 730 days	5,240	3,160	0.603
h = 1095 days	5,279	2,924	0.554

Notes: The bandwidth is varied for both average costs, and $\overline{MC}$ is recomputed from the identity at each one. Coverage shares are held at their Table 6 values, since they come from the point-in-time stock and do not depend on this bandwidth. Point estimates only. The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth; the magnitude of the gap does not.

C.2. How Often the Benefit Ceiling Binds

Table C2 reports the extent to which enrollment spells approach the benefit ceiling on either side of the threshold. This provides the evidence for the contract-equivalence condition in Section 5.4:

although the ceiling changes at age 70 from a household total to an individual entitlement, it is reached by only a negligible share of spells on either side. The change in the unit of the ceiling therefore cannot account for the change in realised claims.

Table C2 — How often does the coverage ceiling bind?

Group	P90	P99	Share $\geq$ 80k	Share $\geq$ 100k	N
Household-spells below 70 (shared 100k ceiling)	20,317	64,226	0.48%	0.09%	288,399
Individual spells 70+ (individual 100k ceiling)	8,225	43,214	0.17%	0.02%	822,009

Notes: Claimed amounts within the analysis sample (spells starting May 2023–May 2024, ages 60–80). Below 70 the NPR 100,000 ceiling is shared at the household level, so totals are summed over household-spell; at and above 70 the ceiling is individual.

Appendix D — Robustness and Inference

D.1. Robustness of the Main Estimates

Table D1 reports the specification checks summarized in Section 6.2: alternative bandwidths, donut and symmetric-donut specifications, a triangular kernel, quadratic polynomials at wider bandwidths, and the MSE-optimal data-driven bandwidth of Calonico, Cattaneo, and Titiunik (2014).

D.2. Randomisation and Inference

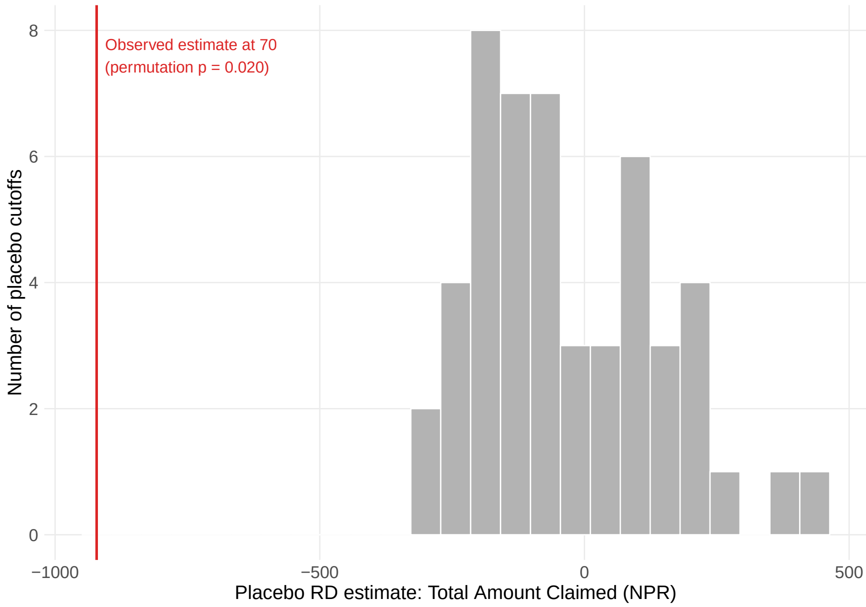
The placebo exercise can be extended beyond the whole-year grid in Figure 2. I re-estimate the discontinuity at 50 placebo cutoffs spaced one-quarter of a year apart across ages 62–68 and 72–78, using only spells on one side of age 70 so that no placebo window contains the true threshold. Treating these estimates as the distribution of discontinuities at ages where no insurance policy changes, the age-70 estimate exceeds every placebo estimate in absolute value for both any claim and total claimed. The resulting permutation p-value is $1/51 \approx 0.02$, the smallest attainable with 50 placebo draws. Figure D1 plots the observed estimate against the placebo distribution; no placebo estimate approaches the magnitude of the age-70 discontinuity.

Table D1 — Robustness of the RD estimates to alternative specifications

	Any Claim	N Claims	Total Claimed (NPR)	Mean Claim (NPR)
	(1)	(2)	(3)	(4)
Panel A: Bandwidth sensitivity (uniform kernel, local linear, no donut)				
$h = 180$ days	-0.053 (0.010)	-0.260 (0.076)	-639.3 (210.2)	-177.5 (48.5)
$h = 312$ days	-0.052 (0.007)	-0.223 (0.057)	-480.0 (158.1)	-135.6 (35.9)
$h = 485$ days	-0.066 (0.006)	-0.471 (0.046)	-921.5 (126.9)	-157.3 (27.5)
$h = 764$ days	-0.072 (0.005)	-0.470 (0.037)	-930.7 (101.6)	-160.1 (21.8)
$h = 1095$ days	-0.086 (0.004)	-0.611 (0.031)	-1,135.0 (84.9)	-171.5 (17.9)
Panel B: Donut (drop the quarter below the cutoff, $-91 \leq rv < 0$)				
$h = 180$ days	-0.046 (0.032)	-0.113 (0.258)	-192.5 (697.0)	6.3 (123.8)
$h = 312$ days	-0.066 (0.013)	-0.393 (0.106)	-857.6 (287.2)	-120.4 (55.9)
$h = 485$ days	-0.083 (0.008)	-0.731 (0.067)	-1,460.8 (181.0)	-195.6 (34.3)
$h = 764$ days	-0.080 (0.006)	-0.580 (0.047)	-1,168.1 (126.2)	-174.7 (25.1)
$h = 1095$ days	-0.094 (0.004)	-0.702 (0.036)	-1,300.1 (98.2)	-179.4 (19.6)
Panel B2: Symmetric donut (drop $\lvert rv \rvert \leq 91$ days)				
$h = 180$ days	-0.007 (0.037)	0.467 (0.291)	981.2 (793.6)	148.0 (149.0)
$h = 312$ days	-0.048 (0.015)	-0.135 (0.116)	-208.0 (317.0)	-32.4 (64.4)
$h = 485$ days	-0.077 (0.009)	-0.679 (0.072)	-1,263.5 (195.4)	-145.6 (38.3)
$h = 764$ days	-0.079 (0.006)	-0.564 (0.049)	-1,088.7 (132.8)	-153.4 (27.0)
$h = 1095$ days	-0.096 (0.005)	-0.719 (0.038)	-1,303.0 (101.8)	-171.2 (20.7)
Panel C: Triangular kernel (local linear)				
$h = 180$ days	-0.057 (0.011)	-0.387 (0.083)	-961.6 (229.5)	-211.0 (55.9)
$h = 312$ days	-0.054 (0.008)	-0.265 (0.062)	-627.9 (172.5)	-161.9 (40.6)
$h = 485$ days	-0.058 (0.007)	-0.339 (0.050)	-692.7 (137.6)	-143.7 (31.5)
$h = 764$ days	-0.068 (0.005)	-0.447 (0.040)	-903.4 (110.9)	-159.3 (24.5)
$h = 1095$ days	-0.076 (0.004)	-0.528 (0.034)	-1,015.3 (92.9)	-162.5 (20.2)
Panel D: Quadratic polynomial at the wider windows (uniform kernel)				
$h = 485$ days	-0.046 (0.009)	-0.132 (0.069)	-342.2 (190.1)	-125.3 (43.8)
$h = 764$ days	-0.062 (0.007)	-0.412 (0.055)	-862.9 (150.6)	-158.1 (33.5)
$h = 1095$ days	-0.061 (0.006)	-0.402 (0.046)	-834.1 (126.3)	-148.5 (27.7)
Panel E: Data-driven bandwidth (rdrobust, uniform, $p = 1$)				
MSE-optimal	-0.059 [0.008]	-0.223 [0.060]	-921.5 [140.8]	-161.6 [23.6]
h (days)	342	312	485	764

Notes: The running variable is days from the seventieth birthday. Every cell is a local linear fit with separate slopes on each side, no covariates and no fixed effects, on the 2023–24 enrolment cycle. Panels A–D report `feols` standard errors clustered on the household in parentheses. Panel E reports the specification of Table 5: the MSE-optimal bandwidth of Calonico, Cattaneo, and Titiunik (2014), selected separately for each outcome and shown in the last row, with robust bias-corrected standard errors in brackets. The selected bandwidth differs by outcome, so no row of Panel A is the main specification for more than one column; Panel E is.

Figure D1 — The age-70 discontinuity lies outside the permutation distribution



Note: The histogram shows RD estimates for total claimed from 50 placebo cutoffs spaced a quarter of a year apart across ages 62–68 and 72–78, each estimated with the specification of Table 5, local linear in age in days with a uniform kernel, no covariates and no fixed effects, at the outcome's own mean-squared-error-optimal bandwidth; the vertical line marks the observed age-70 estimate. The distribution for any claim is analogous. The observed estimate exceeds all 50 placebo estimates in magnitude for both outcomes, so the inclusive permutation p-value, $(1 + \#\{|\hat{\tau}_{\text{plac}}| \geq |\hat{\tau}_{\text{obs}}|\})/(1 + 50) = 1/51 \approx 0.02$, is the smallest attainable with a grid of this size and is the value annotated on the figure.

Inference is robust to the choice of clustering. The baseline specification clusters by household, the level at which enrollment decisions and premiums are determined. Because the running variable takes discrete values, residual dependence across spells sharing the same value may remain (Lee and Card 2008). Table D2 therefore re-estimates standard errors clustering by the running-variable value. Measured in days, the running variable has between 623 and 1,527 distinct values across columns, so the number of clusters is sufficiently large to avoid the few-cluster problem associated with coarser groupings.

Clustering by running-variable value increases the standard errors from 0.007 to 0.009 for any claim, from 0.057 to 0.078 for the number of claims, from 126.9 to 167.1 for total claimed, and from 21.9 to 23.3 for the mean claim. All estimates remain significant at the 1 percent level, with the number-of-claims estimate closest to the threshold.

Table D2 — Inference robustness: clustering by running variable

	Any claim (0/1) (1)	Number of claims (count) (2)	Total claimed (NPR) (3)	Mean claim (NPR) (4)
RD estimate	-0.059	-0.223	-921.5	-161.6
SE, clustered by household	(0.007)	(0.057)	(126.9)	(21.9)
SE, clustered by running variable	(0.009)	(0.078)	(167.1)	(23.3)
Bandwidth (days)	342	312	485	764
Running-variable cells	685	623	971	1,527
Spells	88,148	81,242	127,359	205,305

Notes: The specification is the one used in the main claims table: local linear in age in days with separate slopes on each side of the cutoff, a uniform kernel, no covariates and no fixed effects, each column at its own mean-squared-error-optimal bandwidth. The second row clusters on the household, as the main table does. The third clusters on the value of the running variable, which addresses specification error common to spells sharing a running-variable cell (Lee and Card 2008).

D.3. Strategic Enrolment Timing

If enrollees timed enrollment to coincide with the free-premium threshold by allowing coverage to lapse until age 70, the pool just above the cutoff would be selected on anticipated healthcare needs, making the discontinuity reflect strategic timing rather than price-driven selection. The enrollment-gap distribution provides no evidence of such behavior. The tables below report the distribution of gaps between consecutive spells by age, together with pre-70 claims for enrollees whose gap spans their seventieth birthday relative to all other enrollees. Gaps around age 70 are no longer than elsewhere, while gap-spanners record NPR 8,757 in pre-70 claims compared with NPR 19,354 for other enrollees. Their lower pre-70 costs are the opposite of what strategic timing would predict.

Table D3 — Enrollment gap patterns by age at renewal (%)

Age	No gap	1 quarter	2–3 quarters	4+ quarters
60	75.5	11.0	5.7	7.8
61	75.4	11.1	5.7	7.8
62	75.6	11.0	5.7	7.6
63	76.0	10.8	5.6	7.6
64	75.9	10.9	5.6	7.6
65	76.1	10.8	5.5	7.6
66	75.9	11.0	5.6	7.4
67	76.2	10.8	5.6	7.4
68	76.6	10.7	5.3	7.4
69	76.9	10.7	5.3	7.1
70	**78.8**	**9.5**	**5.1**	**6.7**
71	**91.5**	**3.7**	**2.0**	**2.8**
72	94.6	2.2	1.3	1.9
73	95.9	1.7	1.0	1.5
74	96.5	1.4	0.8	1.3
75	96.9	1.3	0.7	1.1
76	97.0	1.2	0.7	1.1
77	97.1	1.2	0.6	1.1
78	97.2	1.1	0.6	1.1
79	97.2	1.1	0.6	1.1

Table D4 — Pre-70 claims by enrollment transition type

Transition type	N	Mean total claimed (NPR)	Mean no. of claims	% with any claim
Panel A: By gap length				
Continuous (no gap)	88,036	20,925	10.5	68.5
Short gap (< 1 quarter)	7,920	17,669	8.6	72.3
Medium gap (1–2 quarters)	7,973	14,529	7.1	67.8
Long gap (> 2 quarters)	25,344	7,750	3.7	54.1
Panel B: Gap spans 70th birthday				
No	109,682	19,354	—	67.6
Yes	19,591	8,757	—	56.4